

PREDICTIVE ANALYTICS WITH ATHLETICS DATA

Week 7 · Exam 1 review

01

WEEK 1

Describing a distribution



Because many sports statistics have a lower bound of 0 and no technical upper bound, their distributions are most often:

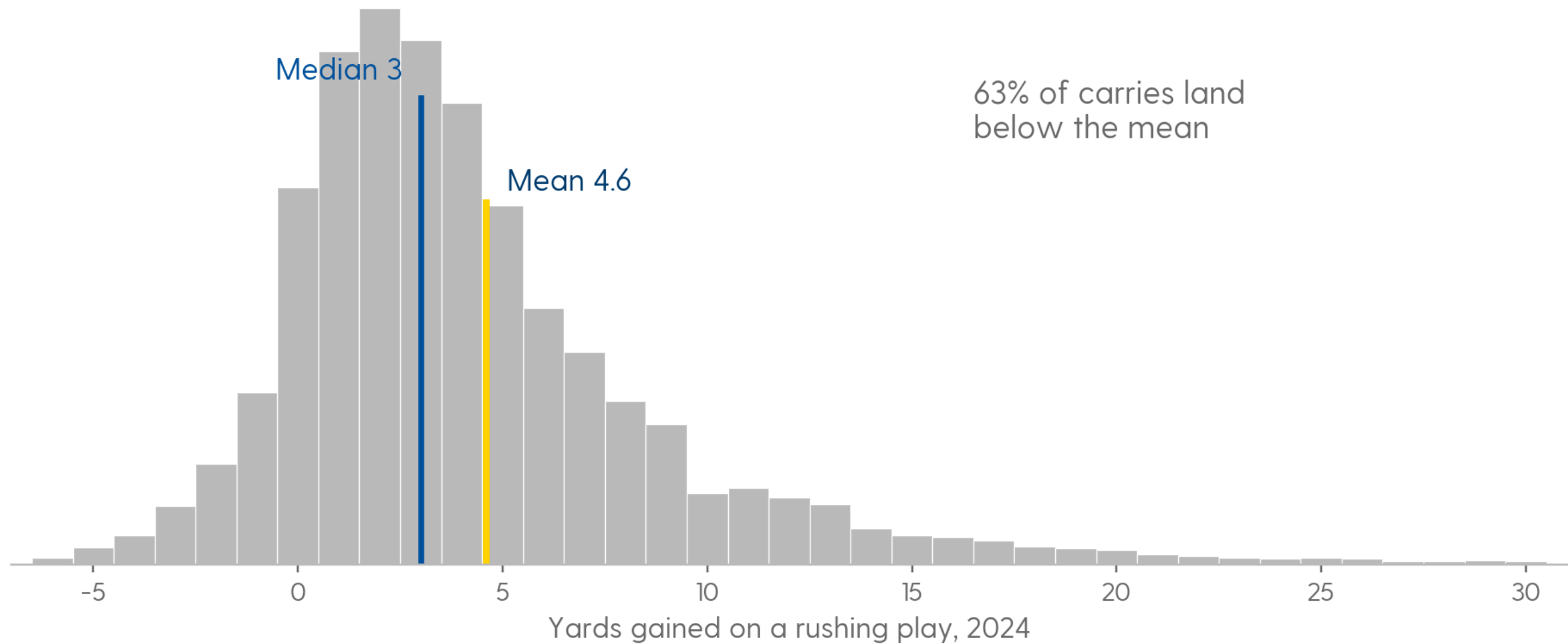
- A** Right skewed.
- B** Bimodal.
- C** Symmetric.
- D** Left skewed.



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- C** Symmetric.
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RUSHING YARDS, 2024



A. Right skewed. Gains pile up near zero and stretch right, so the mean (4.6) sits above the median (3.0).



Kevin Durant, 2010-11, per game: points 27.7 (standard deviation 6.5), rebounds 6.8 (3.3), assists 2.7 (1.4), turnovers 2.8 (1.6). Which statistic has the most variation, based on the coefficient of variation?

- A** Turnovers per game.
- B** Assists per game.
- C** Rebounds per game.
- D** Points per game.



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B Assists per game.

C Rebounds per game.

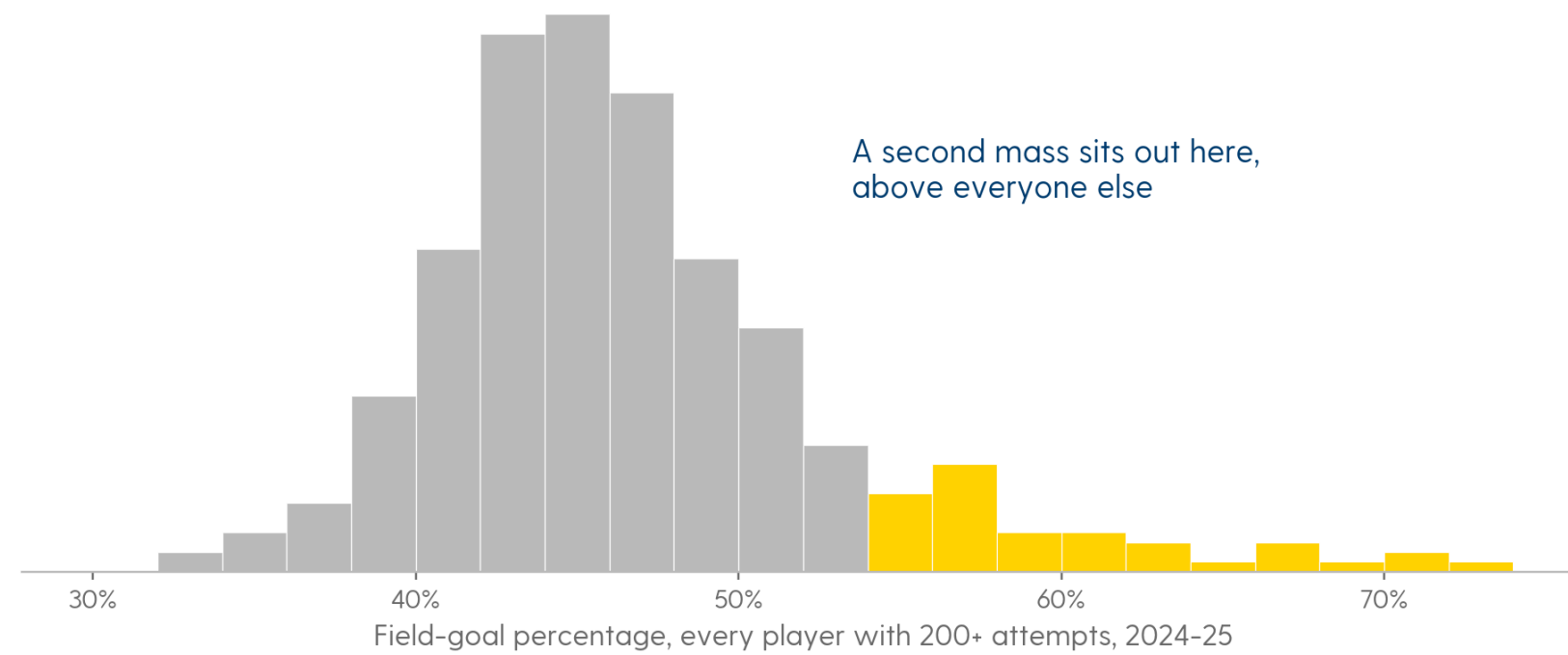
D Points per game.

KEVIN DURANT, 2010-11

STATISTIC	MEAN	STANDARD DEVIATION	COEFFICIENT OF VARIATION
Points per game	27.7	6.5	0.235
Rebounds per game	6.8	3.3	0.485
Assists per game	2.7	1.4	0.519
Turnovers per game	2.8	1.6	0.571

A. Turnovers. Points has the largest standard deviation and the smallest coefficient of variation.

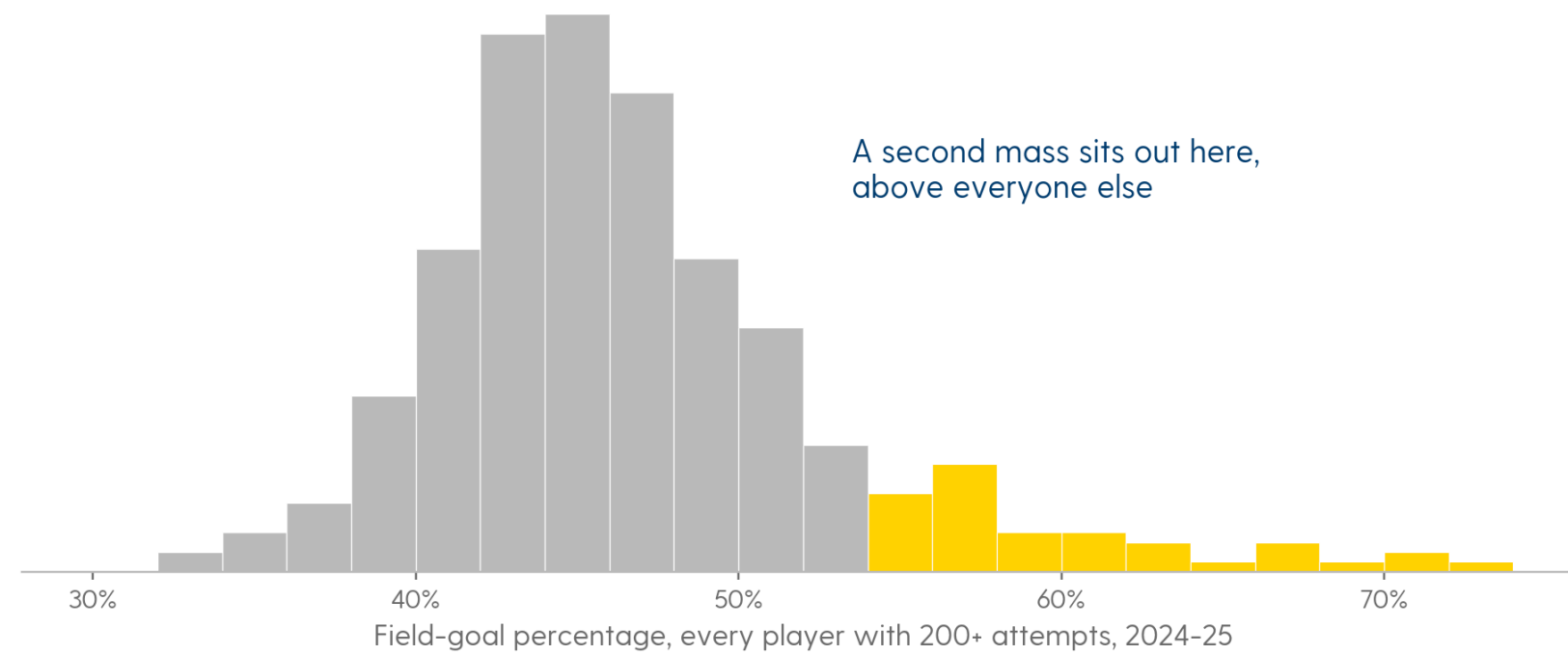
WEEK 1, QUESTION 3



Bimodal distributions are often an indicator that your data has multiple:

- A** Means.
- B** Subclasses.
- C** Skews.
- D** Standard deviations.

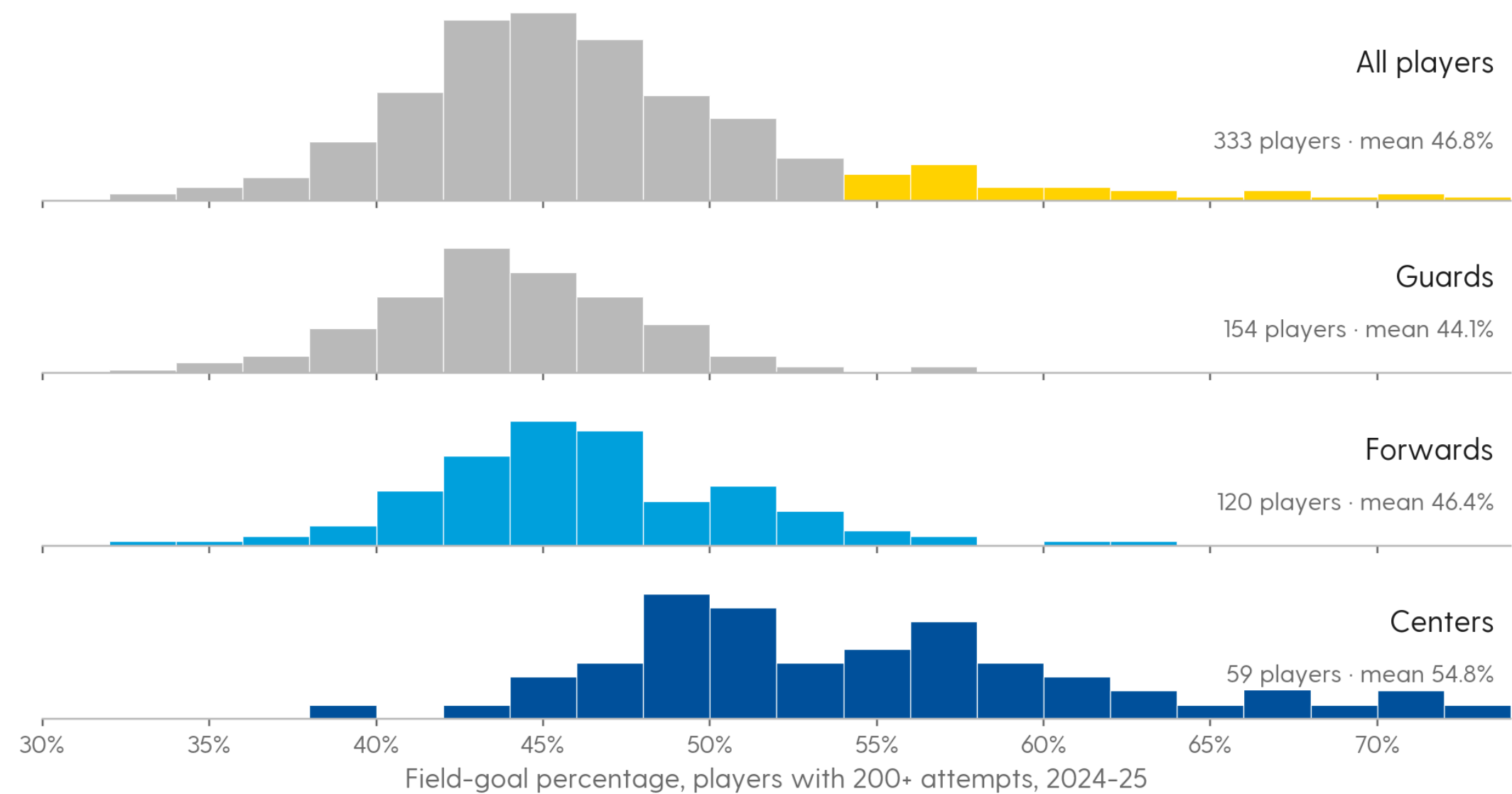
WEEK 1, QUESTION 3



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- B Subclasses.**
- C Skews.
- D Standard deviations.

FIELD-GOAL PERCENTAGE BY POSITION, 2024-25



B. Subclasses. Two peaks usually mean two kinds of player in one file. Split them and each group has one peak.



The maximum standardized entropy for a binomial distribution, such as a football team's situational pass/run distribution: Select all that apply.

- A** Is 1.
- B** Occurs when the pass/run split is 100%/0% or 0%/100%.
- C** Is $\ln(2)$.
- D** Occurs when the pass/run split is 50%/50%.



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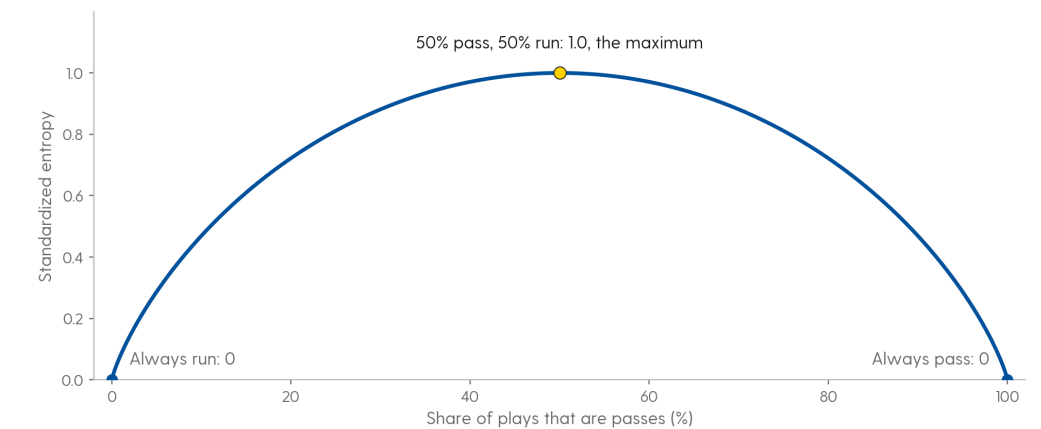
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STANDARDIZED ENTROPY OF A PASS AND RUN SPLIT

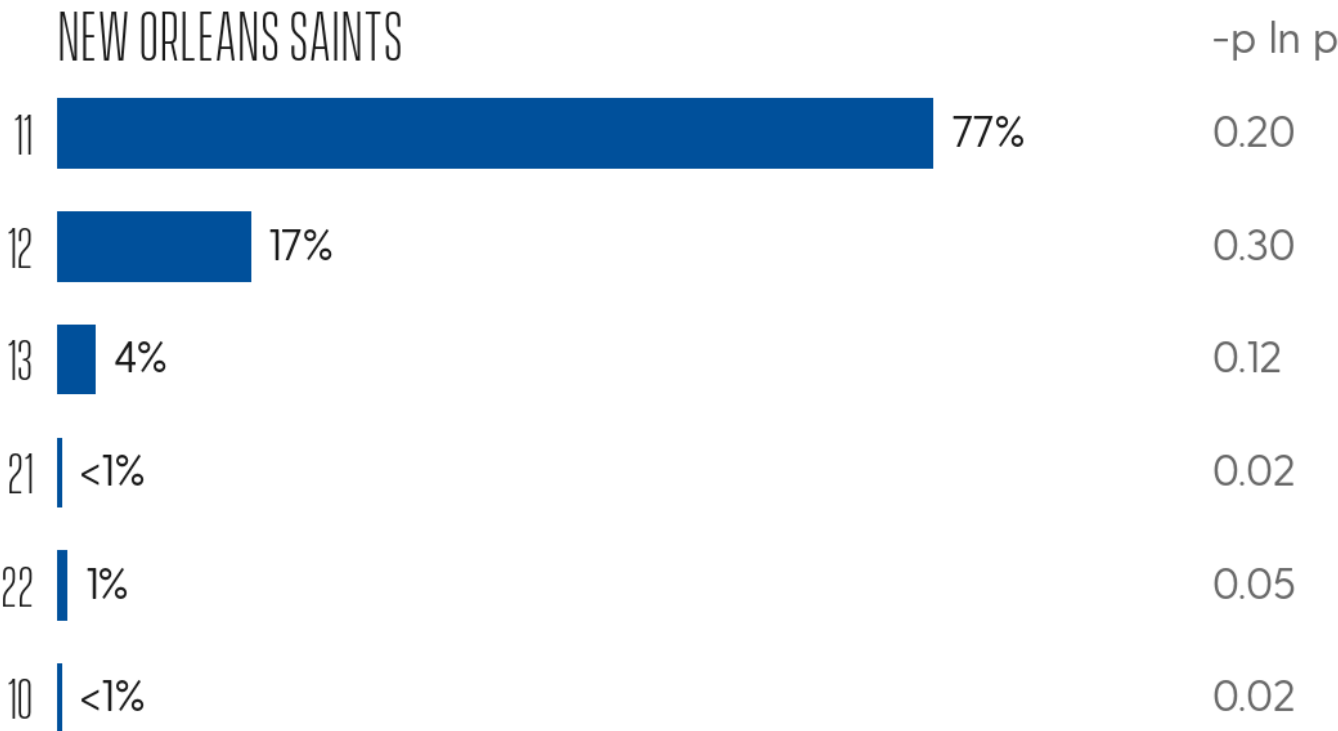
$$\text{Standardized} = \frac{-[p \ln p + (1 - p) \ln(1 - p)]}{\ln 2}$$

$$\text{Even split} = \frac{-[0.5 \ln 0.5 + 0.5 \ln 0.5]}{\ln 2} = 0.69 / 0.69 = 1$$

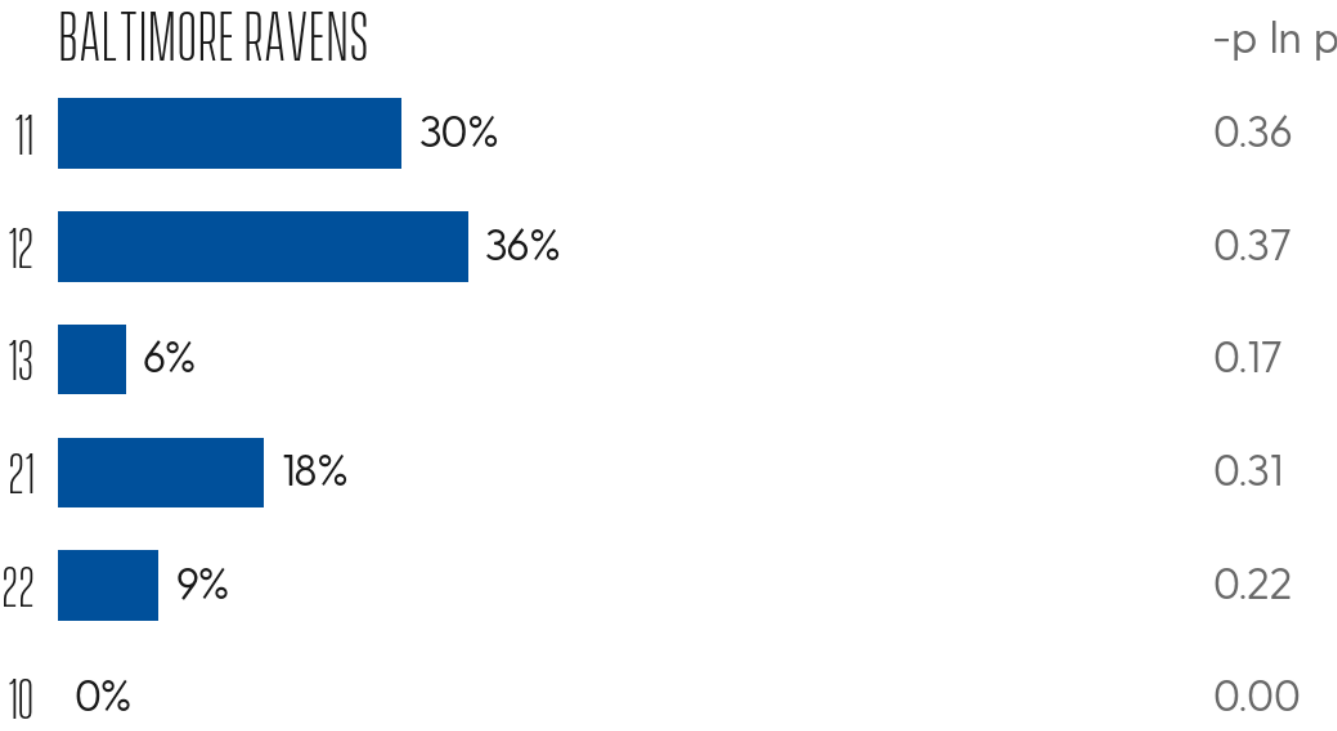
A AND D. THE MAXIMUM IS 1, AT AN EVEN SPLIT. LN(2) IS THE UNSTANDARDIZED MAXIMUM, BEFORE DIVIDING BY LN(N).



OFFENSIVE PERSONNEL, 2025



$H = 0.20 + 0.30 + 0.12 + 0.02 + 0.05 + 0.02 = 0.71$
 $\text{Standardized} = 0.71 \div \ln 6 = 0.71 \div 1.79 = 0.40$



$H = 0.36 + 0.37 + 0.17 + 0.31 + 0.22 + 0.00 = 1.43$
 $\text{Standardized} = 1.43 \div \ln 6 = 1.43 \div 1.79 = 0.80$

The Saints lived in 11 and had the lowest standardized entropy in the league. The Ravens spread snaps across five groupings and had the highest. All six used equally would be $\ln 6 / \ln 6 = 1$.

nflverse play participation (FTN charting), 2025 season.

02

WEEK 2

Probability and inference



Shai Gilgeous-Alexander made 89.8% of his free throws last season. The following formula returns the probability that he: =BINOM.DIST(8, 10, 0.898, TRUE)

- A** Misses exactly eight of ten attempts.
- B** Makes eight or more of ten attempts.
- C** Makes exactly eight of ten attempts.
- D** Makes eight or fewer of ten attempts.



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THE FOURTH ARGUMENT OF BINOM.DIST

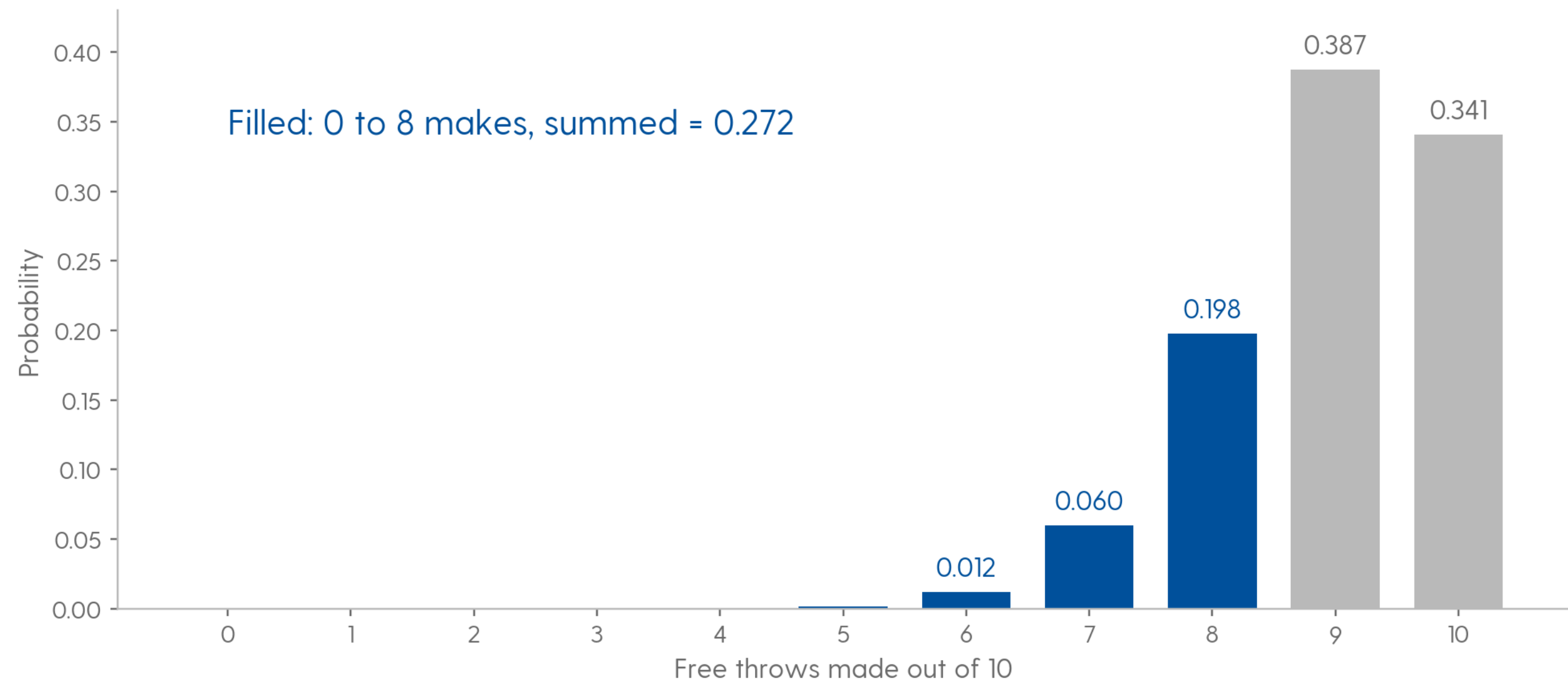
=BINOM.DIST(8, 10, 0.898, TRUE) = P(8 or fewer) = 0.272

=BINOM.DIST(8, 10, 0.898, FALSE) = P(exactly 8) = 0.198

1 – BINOM.DIST(7, 10, 0.898, TRUE) = P(8 or more) = 0.926

D. TRUE ADDS UP EVERY OUTCOME FROM 0 TO 8.

SHAI GILGEOUS-ALEXANDER, TEN FREE THROWS



TRUE fills in every bar from 0 to 8. Most of his probability sits on 9 and 10, so eight or fewer is only about one time in four.



The Washington Capitals are favored in 50 of their 82 games this season. How many games are they expected to win?

- A** Not enough information is given.
- B** Exactly 50 games.
- C** More than 50 games.
- D** Fewer than 50 games.



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A. NOT ENOUGH INFORMATION.

Favored means a win probability above 50%, by an unknown amount, in 50 games. Expected wins is the sum of all 82 win probabilities, and none of them were given.

If they were 60% favorites in 25 games, 70% favorites in 25 games, and had a 40% chance to win as underdogs in the remaining 32: $25 \times 0.60 + 25 \times 0.70 + 32 \times 0.40 = 15 + 17.5 + 12.8 = 45.3$ expected wins.



Giannis Antetokounmpo shot 61.7% from the free throw line last season. He steps to the line for two. Treating the attempts as independent, what is the most likely outcome?

- A** He misses both attempts.
- B** He makes both attempts.
- C** Not enough information is given.
- D** He misses one or both attempts.



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GIANNIS ANTETOKOUNMPO, 2024-25

OUTCOME	MATH	PROBABILITY
Makes both	0.617×0.617	38.0%
Makes exactly one	$2 \times 0.617 \times 0.383$	47.3%
Misses both	0.383×0.383	14.7%
Misses one or both	$1 - 0.617 \times 0.617$	62.0%

D. He made 61.7% of his free throws. Makes both is the likeliest single outcome, but missing one or both groups two outcomes and is larger.



Aaron Judge got 179 hits in 541 at bats last season, a .331 average, with a standard error of 2.0 percentage points. This means:

- A** From last season's at bats, we can estimate with 95% confidence that his true rate of getting a hit lies between .311 and .351.
- B** From last season's at bats, we can estimate with 95% confidence that next season he will hit between .291 and .370.
- C** From last season's at bats, we can estimate with 95% confidence that his true rate of getting a hit lies between .291 and .370.
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- D** From last season's at bats, we can estimate with 95% confidence that next season he will hit between .311 and .351.

AARON JUDGE, 179 HITS IN 541 AT BATS

95%: $.331 \pm 1.96 \times .020 = 0.291$ to 0.370

68%: $.331 \pm 1 \times .020 = 0.311$ to 0.351

Both describe his true rate in the season we watched, not next season.

C. TWO STANDARD ERRORS, AND HIS UNDERLYING RATE RATHER THAN A FORECAST.



Giannis Antetokounmpo shot 61.7% from the free throw line last season. A shooter at that rate who has taken 100 attempts has a margin of error of about 9.5 percentage points. Roughly how many attempts would it take for that margin of error to fall to about 4.8 points?

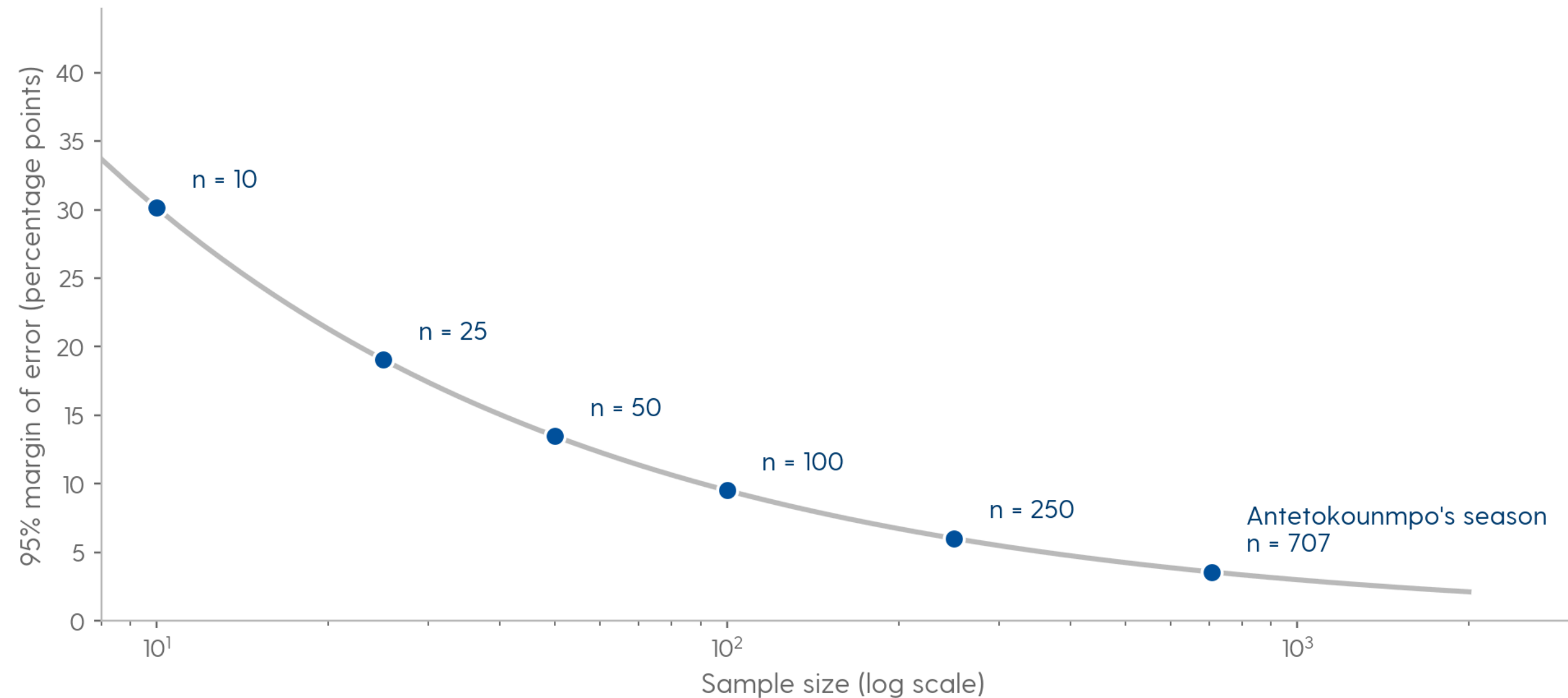
- A** About 110, because the margin of error falls by about a point per additional attempt.
- B** About 400, because the margin of error shrinks with the square root of the sample size.
- C** About 200, because doubling the sample halves the margin of error.
- D** There is no number of attempts that would help, because the rate itself is uncertain.



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MARGIN OF ERROR BY SAMPLE SIZE



B. About 400. The margin shrinks with the square root of n : 9.5 points at 100 attempts, 4.8 at 400. Four times the attempts halves it.



In 2023, E. McPherson made 26 of 31 field goals, or 83.9%, and the league made 81.4% of attempts from the distances he was sent out from. The league made 85.8% of all attempts that season. He ranked 19th of 31 qualifying kickers on raw make rate and 12th once adjusted. Which statement describes what the adjustment did?

- A** It removed his longest attempts from the calculation, which raised his rate.
- B** It compared him against the league's overall make rate of 85.8%, and he was 1.9 points worse than that.
- C** It corrected his make rate for the small number of attempts he took.
- D** It compared him against what an average kicker would have made from his own attempts, put the 2.5 point difference back on the league's scale, and gave him an adjusted make rate of 88.3%.



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E. MCPHERSON, 2023

Made 83.9% of his 31 attempts

An average kicker from his distances 81.3%

Over expected +2.5 points

Adjusted = league 85.8% + 2.5 = 88.3%

D. RAW RANK 19, ADJUSTED RANK 12. HIS ATTEMPTS AVERAGED 42.5 YARDS, THE LONGEST OF ANY QUALIFYING KICKER.

03

WEEK 3

Bayes' theorem



Which of the following best describes a prior probability in Bayesian statistics?

- A** The probability of the data given a hypothesis
- B** The probability of a hypothesis before observing data
- C** The probability of a hypothesis after observing data
- D** The probability of a random sample being representative



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BAYESIAN DEFINITIONS

	WHAT IT MEANS
Prior	What you believed before you saw the data
Likelihood	How probable the data is, under each hypothesis
Posterior	What you believe after you have seen the data

B. A is the likelihood and C is the posterior.

THE SIDEWALK QUARTER

My best guess is that 1 in 100,000 coins on the ground is weighted, so a 99.999% chance the coin is fair

3.125% if fair, 100% if weighted

99.97% fair



Which expression reflects Bayes' theorem?

- A** $P(H \mid D) = P(H) / P(D \mid H)$
- B** $P(H \mid D) = P(H) \times P(D \mid H) / P(D)$
- C** $P(D \mid H) = P(H \mid D) \times P(H) / P(D)$
- D** $P(D \mid H) = P(D) / P(H \mid D)$



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- D** $P(D \mid H) = P(D) / P(H \mid D)$

BAYES' THEOREM

$$\text{posterior} \quad P(H \mid D) = \frac{\text{prior} \quad \text{likelihood} \quad P(H) \times P(D \mid H)}{P(D)}$$

$$P(\text{fair} \mid \text{five heads}) = \frac{\text{prior} \quad \text{likelihood} \quad 0.99999 \times 0.03125}{\text{prior} \quad \text{likelihood} \quad 0.99999 \times 0.03125 + 0.00001 \times 1} = 0.9997$$

B. PRIOR TIMES LIKELIHOOD, OVER THE PROBABILITY OF THE DATA.



A QB drafted in the 7th round threw for 5 TDs and 0 INTs in his first game. The fans are saying that the team drafted a hall-of-famer. Which question is the Bayesian one?

- A** Given that he threw for 5 TDs and 0 INTs in his first game, what is the probability he is a hall-of-famer?
- B** Given that he is a hall-of-famer, what is the probability he'd throw for 5 TDs and 0 INTs in his first game?
- C** What is the probability a hall-of-fame QB throws for 5 TDs and 0 INTs in a game?
- D** Is 5 TDs and 0 INTs a statistically significant departure from the average rookie start?



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HYPOTHESIS GIVEN DATA

hypothesis

data

$P(\text{hall-of-famer} \mid 5 \text{ TDs}, 0 \text{ INTs})$

The posterior. The Bayesian question.

data

hypothesis

$P(5 \text{ TDs}, 0 \text{ INTs} \mid \text{hall-of-famer})$

The likelihood. What a p-value computes, and what B, C and D ask.

A. THE HYPOTHESIS IS ON THE LEFT OF THE BAR AND THE DATA ON THE RIGHT.



The chance a fair coin turns up 5 heads in a row is 0.03125. You found the coin on the sidewalk and your friend found one in a magic shop and both got five heads in a row. Which statement is right?

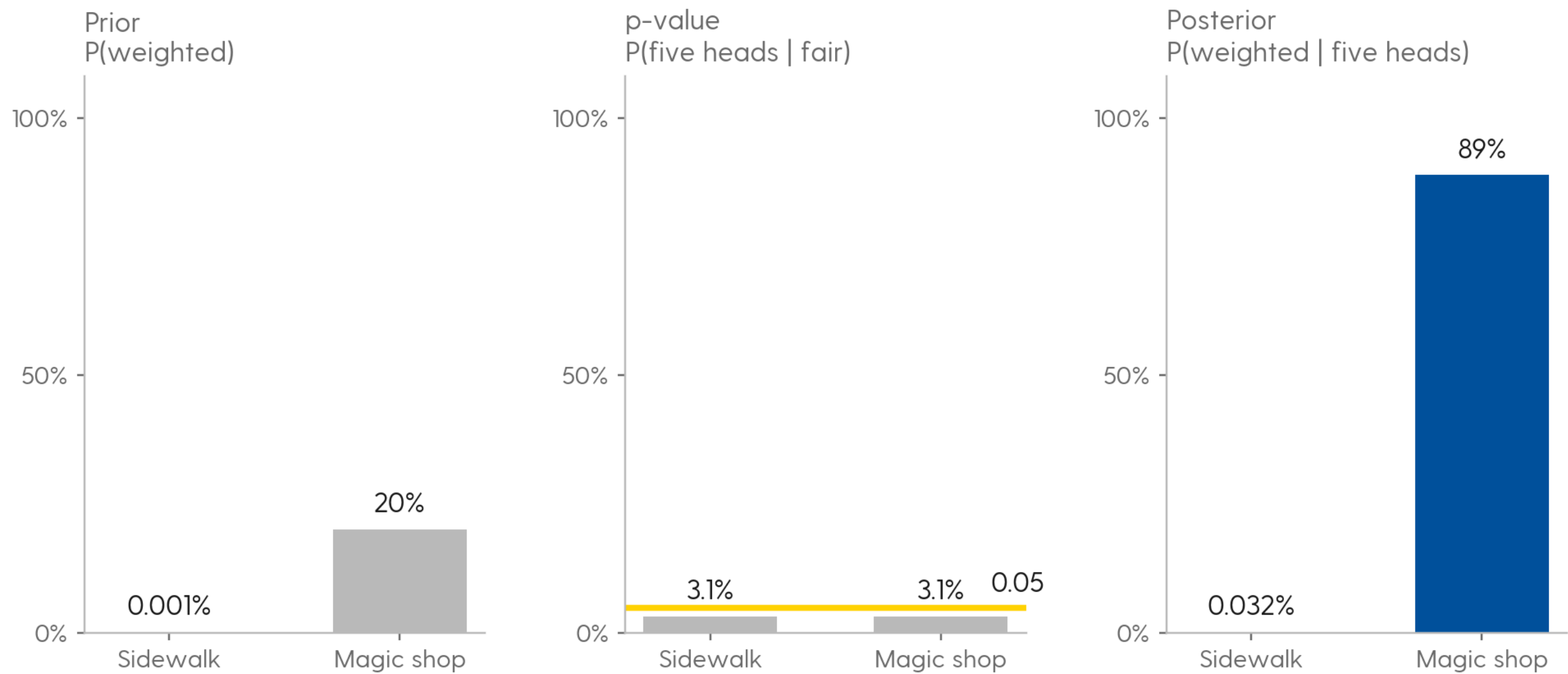
- A** The p-value is the same for both coins.
- B** The p-value is higher for the magic shop coin, because weighted coins are more common there.
- C** The posterior probability that the coin is weighted is the same for both, because the evidence is the same.
- D** Both coins are significantly weighted at the 0.05 level, so both should be treated as weighted.



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THE P-VALUE AND THE POSTERIOR



A. The p-value is 0.03125 in both places. The prior is what differs, and only the posterior can hold it.

SIDEWALK AND MAGIC SHOP

$$\text{Sidewalk: } P(\text{fair} \mid \text{five heads}) = \frac{\overset{\text{prior}}{0.99999} \times \overset{\text{likelihood}}{0.03125}}{0.99999 \times 0.03125 + 0.00001 \times 1} = 0.9997$$

$$\text{Magic shop: } P(\text{fair} \mid \text{five heads}) = \frac{\overset{\text{prior}}{0.80} \times \overset{\text{likelihood}}{0.03125}}{0.80 \times 0.03125 + 0.20 \times 1} = 0.1111$$

SAME EVIDENCE. THE POSTERIOR MOVED FROM 99.97% FAIR TO 11.1% FAIR.



You went into the year believing that there was a 75% chance that rookie quarterback, Fernando Mendoza, would be a "bust". He performs well over the first five games. You calculate that his performance only had a 10% chance of occurring if he is a "bust", but had a 50% chance of occurring if he wasn't a bust. Using your prior and new data in Bayes' Theorem, what is the likelihood that Fernando Mendoza is still a bust?

-
- A** 37.5%
 - B** 7.5%
 - C** 12.5%
 - D** 10.0%



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FERNANDO MENDOZA, 100 ROOKIES

100 rookie quarterbacks, split by the prior



100 rookie quarterbacks who are a bust: 10 play well



100 rookie quarterbacks who are not a bust: 50 play well



Played well over five games (75 x 0.1 and 25 x 0.5)



7.5 of 20 are bust: 37.5%

A. Of the 20 who played well, 7.5 are busts: 37.5%.

BAYES' THEOREM ON FIVE GAMES

$$P(\text{bust} \mid \text{played well}) = \frac{P(\text{bust}) \times P(\text{played well} \mid \text{bust})}{P(\text{played well})}$$

$$= \frac{\begin{matrix} \text{prior} & \text{likelihood} \\ 0.75 \times & 0.10 \end{matrix}}{\begin{matrix} \text{prior} & \text{likelihood} \\ 0.75 \times 0.10 + & 0.25 \times 0.50 \end{matrix}} = 0.075 / 0.200 = 0.375$$

B is the numerator alone. C is the other bucket, 0.25×0.50 . D is the likelihood reported as the posterior.

FIVE GOOD GAMES CUT THE BUST PROBABILITY IN HALF, FROM 75% TO 37.5%. HE IS NOW MORE LIKELY NOT A BUST, AT 62.5%.

04

WEEK 4

Descriptive and probabilistic statistics



Which two of these descriptors are key attributes of Dean Oliver's "Four Factors" that make it effective at explaining the results of a game?

- A** Predictive
- B** Mutually exclusive
- C** Exhaustive
- D** Probabilistic



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DEAN OLIVER'S FOUR FACTORS

FACTOR		WEIGHT
eFG%	Shooting	40%
TO%	Turnovers	25%
ORB%	Rebounding	20%
FTAr	Free throws	15%

B and C. Exhaustive and mutually exclusive: every possession ends in a way one of the four describes, and no possession counts toward two.



What is the primary outcome coaches should optimize for in a soccer game?

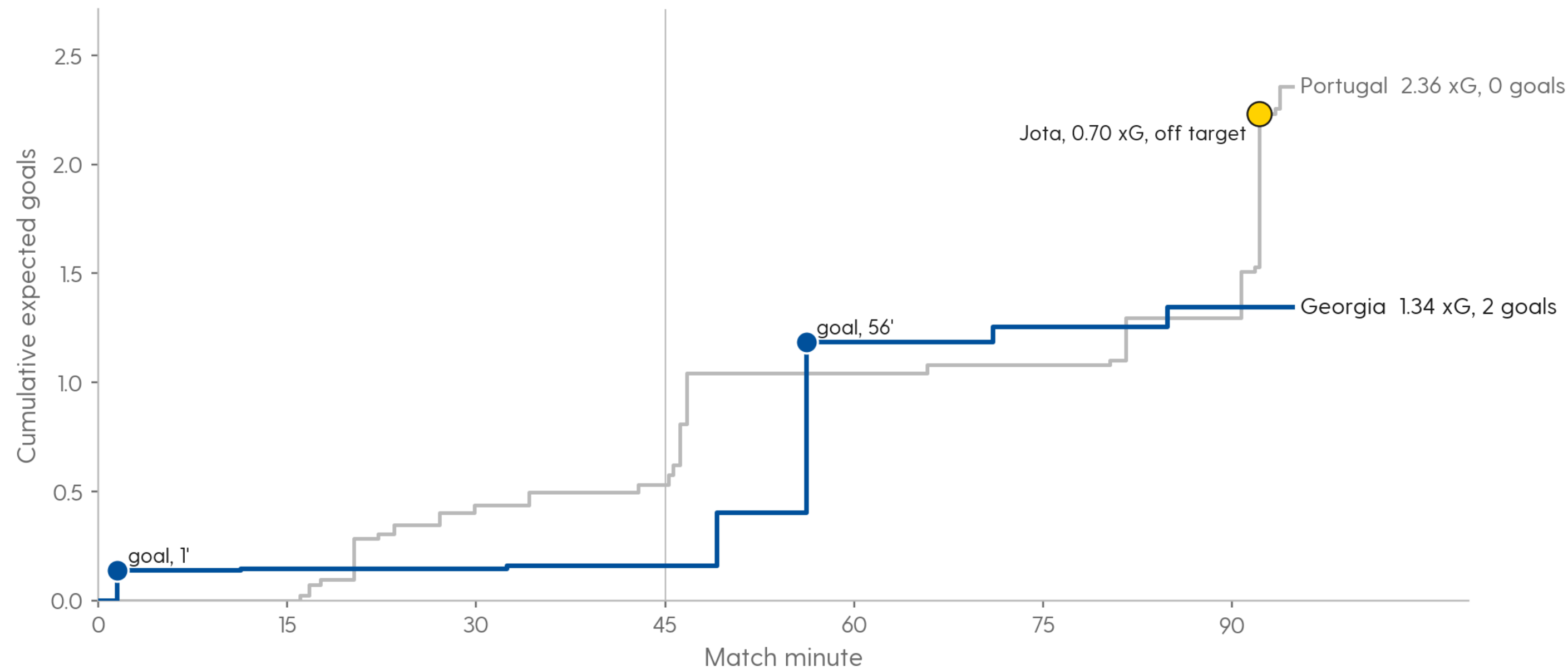
- A** Expected goal differential
- B** Expected goals
- C** Win probability
- D** Actual goal differential



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- B** Expected goals
- C** **Win probability**
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GEORGIA 2, PORTUGAL 0, EURO 2024



C. Win probability. Portugal won the expected goals 2.36 to 1.34 and lost the game.



In basketball, "Points per Possession", an "efficiency statistic", is

- A** a predictive statistic.
- B** a probabilistic statistic.
- C** a descriptive statistic.
- D** None of the above.



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- A** a predictive statistic.
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- C** **a descriptive statistic.**
- D** None of the above.

C. DESCRIPTIVE. IT SUMMARIZES WHAT ALREADY HAPPENED.

DESCRIPTIVE

Points per possession, slugging, box score totals.

PROBABILISTIC

Win probability, expected points: a chance attached to a situation.

PREDICTIVE

A forecast of something that has not happened yet.



When might a team be expected to deviate from optimizing for score differential?

- A** When they are losing by a large margin.
- B** When they are winning by a large margin.
- C** In both of the above situations.
- D** In none of the above situations.



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C. BOTH. FAR AHEAD OR FAR BEHIND, A TEAM PLAYS FOR WIN PROBABILITY, NOT MARGIN.

WINNING BIG

Run the ball, burn clock, take the safe play.

LOSING BIG

Take on variance: fourth downs, onside kicks, early thirds.



Slugging percentage, calculated as $(1 \times 1B + 2 \times 2B + 3 \times 3B + 4 \times HR) / AB$, is a(n)

- A** Counting statistic
- B** Probabilistic statistic
- C** Atomic unit
- D** Weighted efficiency statistic



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- D** **Weighted efficiency statistic**

SLUGGING PERCENTAGE

$$SLG = (1 \times 1B + 2 \times 2B + 3 \times 3B + 4 \times HR) / AB$$

Weights on each kind of hit, divided by the chances he had.

D. A WEIGHTED EFFICIENCY STATISTIC.



In Super Bowl LI, Julian Edelman's overtime catch had a WPA of +0.165. Across the game's 169 plays, WPA had a mean of 0.0013 and a standard deviation of 0.0324. In Excel, `=STANDARDIZE(0.165, 0.0013, 0.0324)` returns 5.0. This number is:

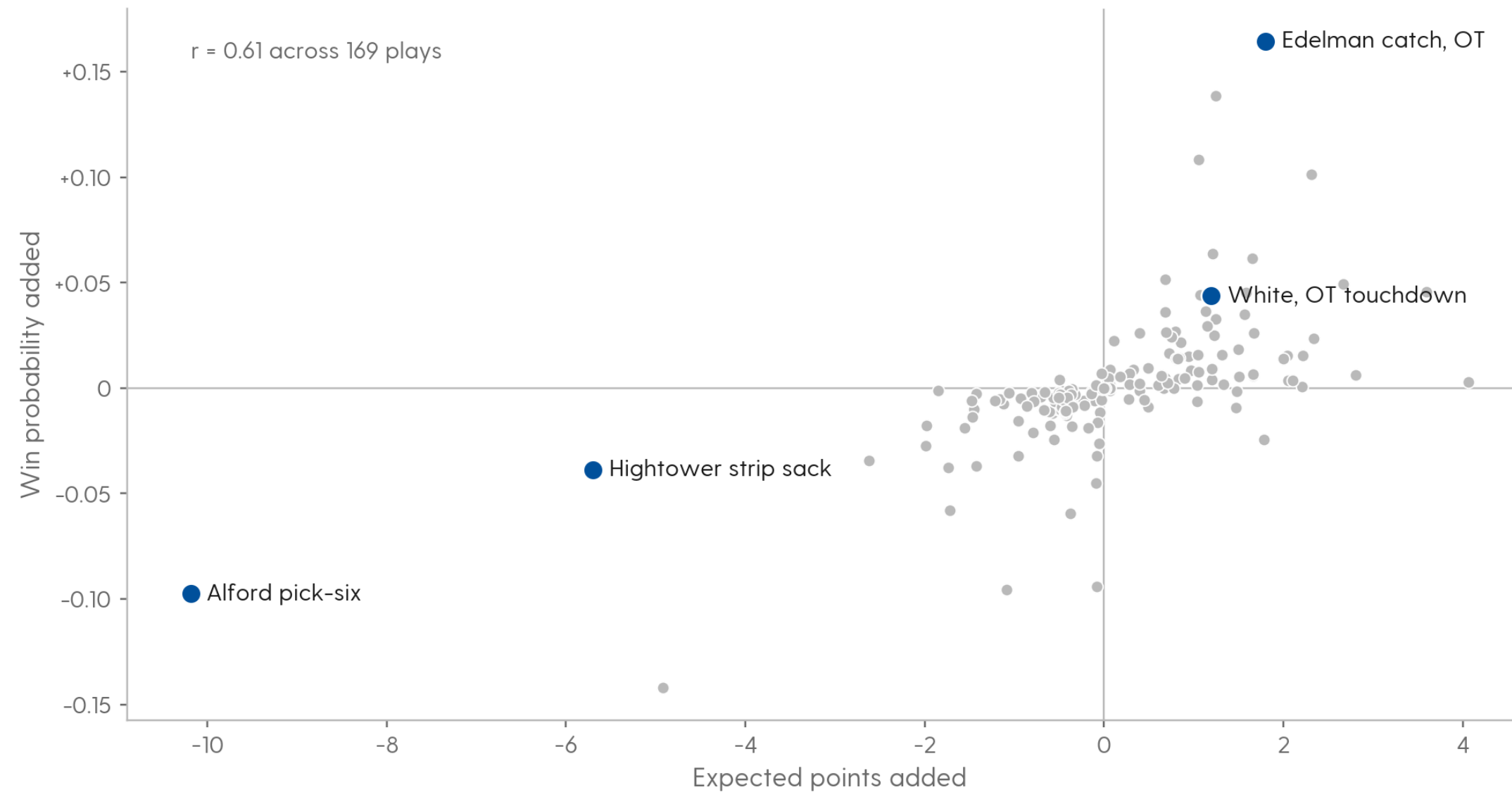
- A** How many standard deviations above the average play this one was, by win probability added.
- B** The correlation between EPA and WPA for this play.
- C** The probability that a play adds at least 0.165 of win probability.
- D** The play's EPA, on the WPA scale.



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- D** The play's EPA, on the WPA scale.

SUPER BOWL LI, EVERY PLAY



A. Edelman's catch was 5.0 standard deviations above the average play by win probability added, and 1.2 by expected points added.

05

WEEK 5

Regression



You regress second-half wins on first-half wins across 270 MLB team-seasons. Excel reports a coefficient of 0.63 on first-half wins with a p-value below 0.0001. The p-value is:

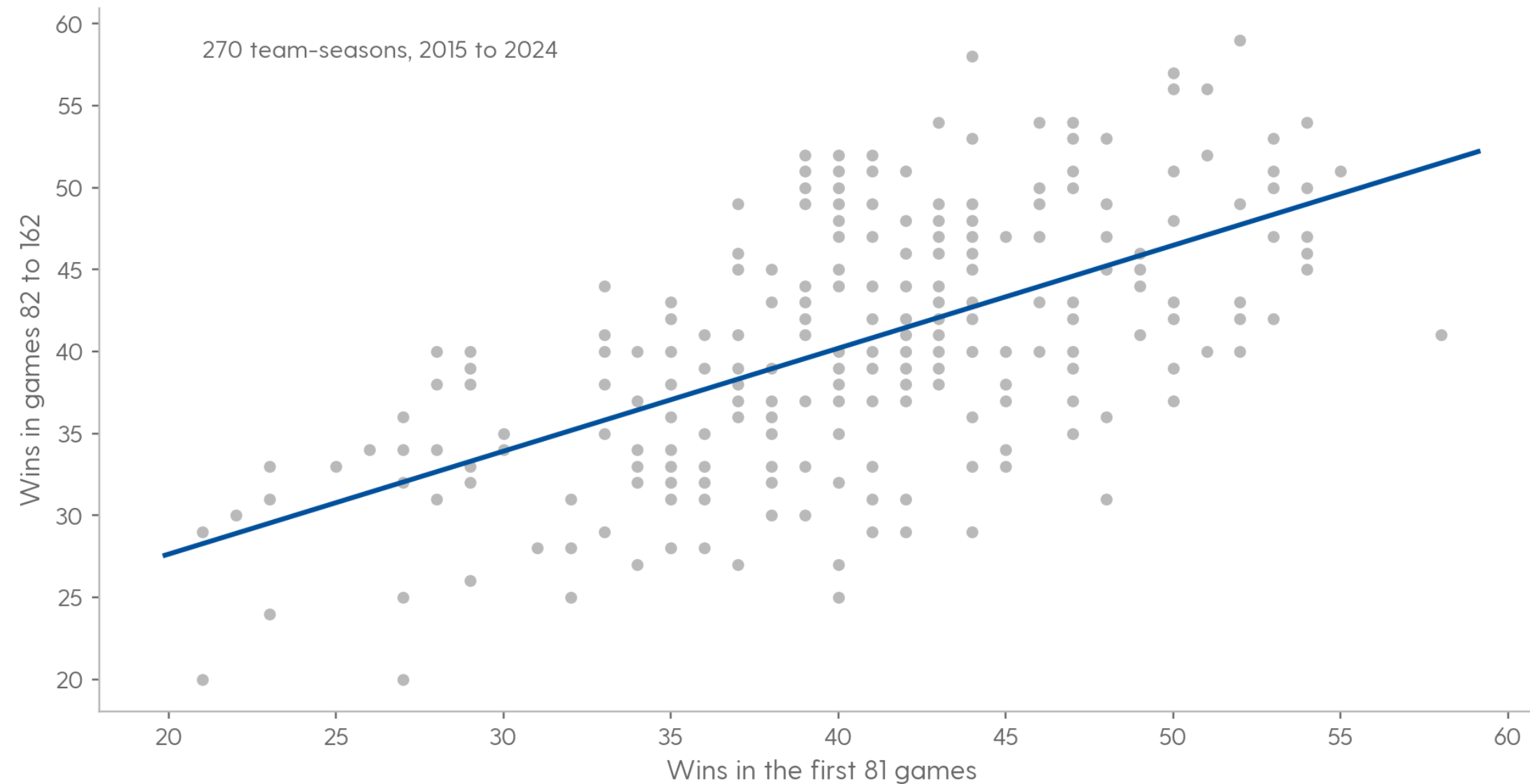
- A** The probability of seeing a slope at least this far from zero if first-half wins had no linear relationship with second-half wins.
- B** The probability that first-half wins cause second-half wins.
- C** The probability that the true slope is 0.63.
- D** The share of second-half wins explained by first-half wins.



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FIRST-HALF AND SECOND-HALF WINS



A. How surprising a slope this far from zero would be if there were no relationship. Not causation, not R squared, not the chance the slope is 0.63.



In the same output, the 95% interval on the first-half wins coefficient runs from 0.52 to 0.73. Which statement is right?

- A** The slope is significant because 0.63 is inside the interval.
- B** 95% of teams' second-half wins fall between 0.52 and 0.73 times their first-half wins.
- C** If we drew many new samples of team-seasons and built this interval each time, about 95 in 100 of those intervals would contain the true slope.
- D** There is a 95% chance the slope in a new sample lands between 0.52 and 0.73.



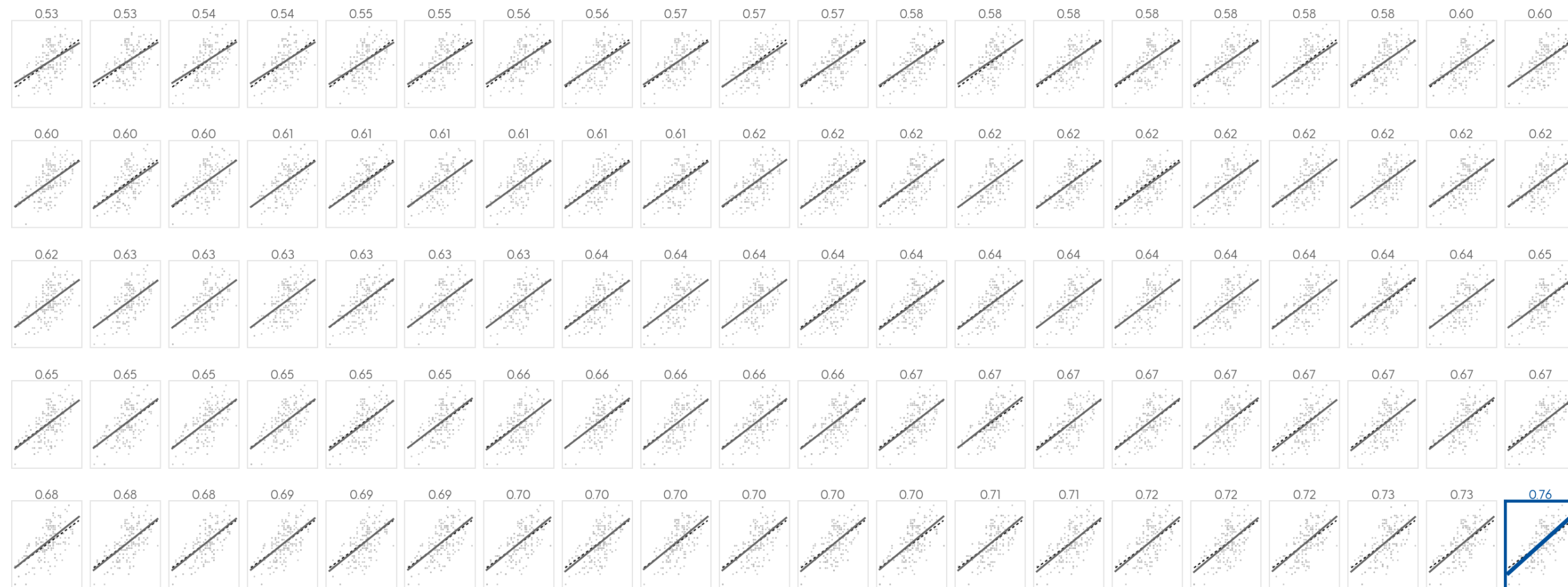
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- D** There is a 95% chance the slope in a new sample lands between 0.52 and 0.73.

C. 95% DESCRIBES THE METHOD.

The 95% is about how often this recipe works, not about this one interval. Pull a new set of seasons, build the interval the same way, and do that 100 times: about 95 of the intervals catch the true slope. Our one interval either caught it or missed it, and we can't tell which.

RESAMPLING THE 270 SEASONS



Our 270 seasons are one set of the seasons that could have happened. Each panel resamples them with replacement and refits, sorted from weakest slope to strongest; the dashed line is our real fit. The slope moves every time, so 0.63 is an estimate, not the answer. Only the blue panel falls outside the output's interval of 0.52 to 0.73.

MLB team-seasons 2015 to 2024 without 2020, first and second halves of 81 games. 100 resampled sets of 270.



Pool all 178,921 pairs of consecutive games by qualified hitters from 2015 through 2024 and hits in one game correlate with hits in the next at $r = 0.027$, p below 0.0001, R squared 0.071%. Subtract each hitter's own season average from both of his games first and the same pairs give $r = 0.005$, $p = 0.044$, R squared 0.002%. The right reading is:

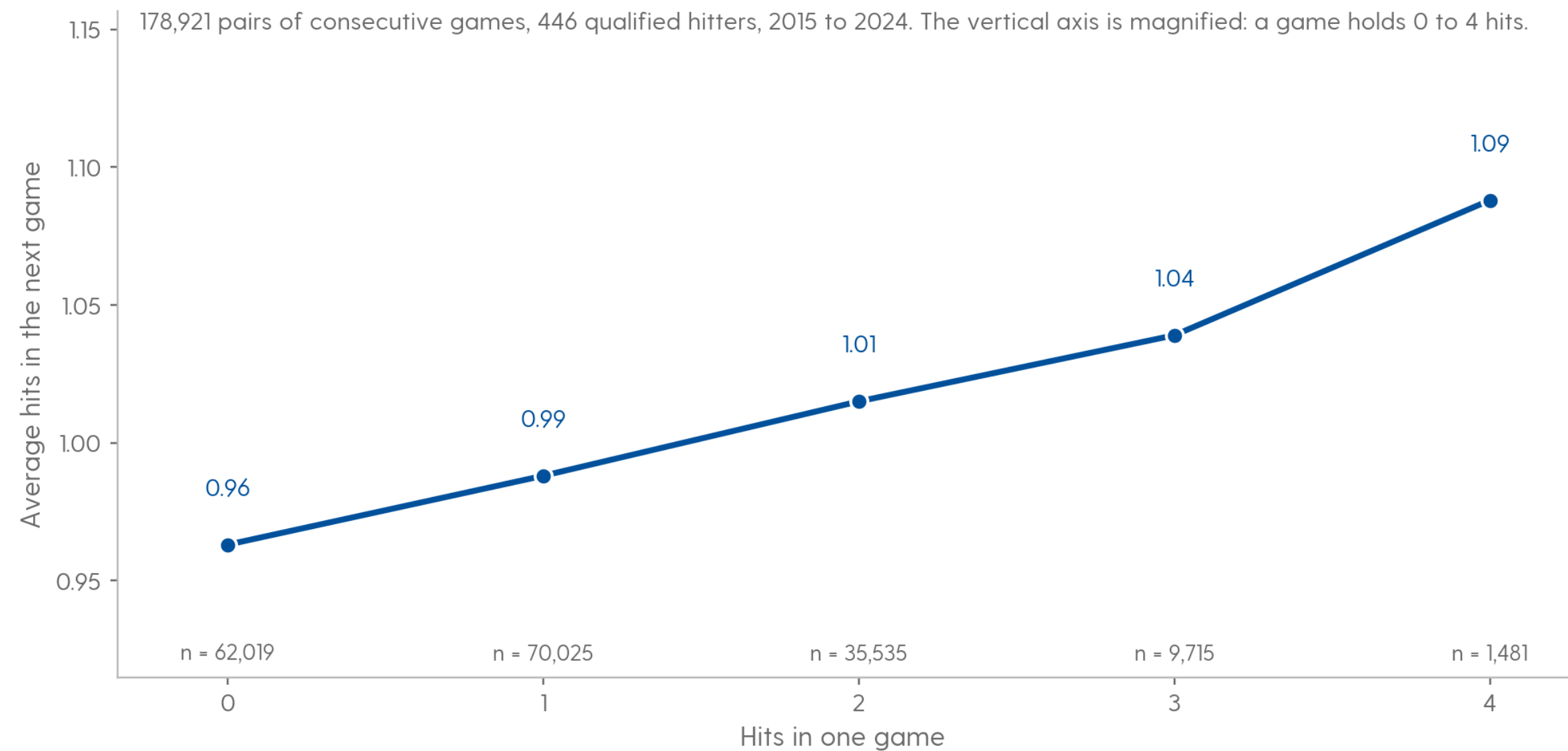
-
- A** The pooled result is not significant because R squared is so small.
 - B** The second number clears 0.05, so a hitter repeating himself is now established.
 - C** The pooled correlation was mostly the gap between good and weak hitters, and what is left is too small to act on.
 - D** Hitters are streaky, and a two-hit game should raise what you expect tomorrow.



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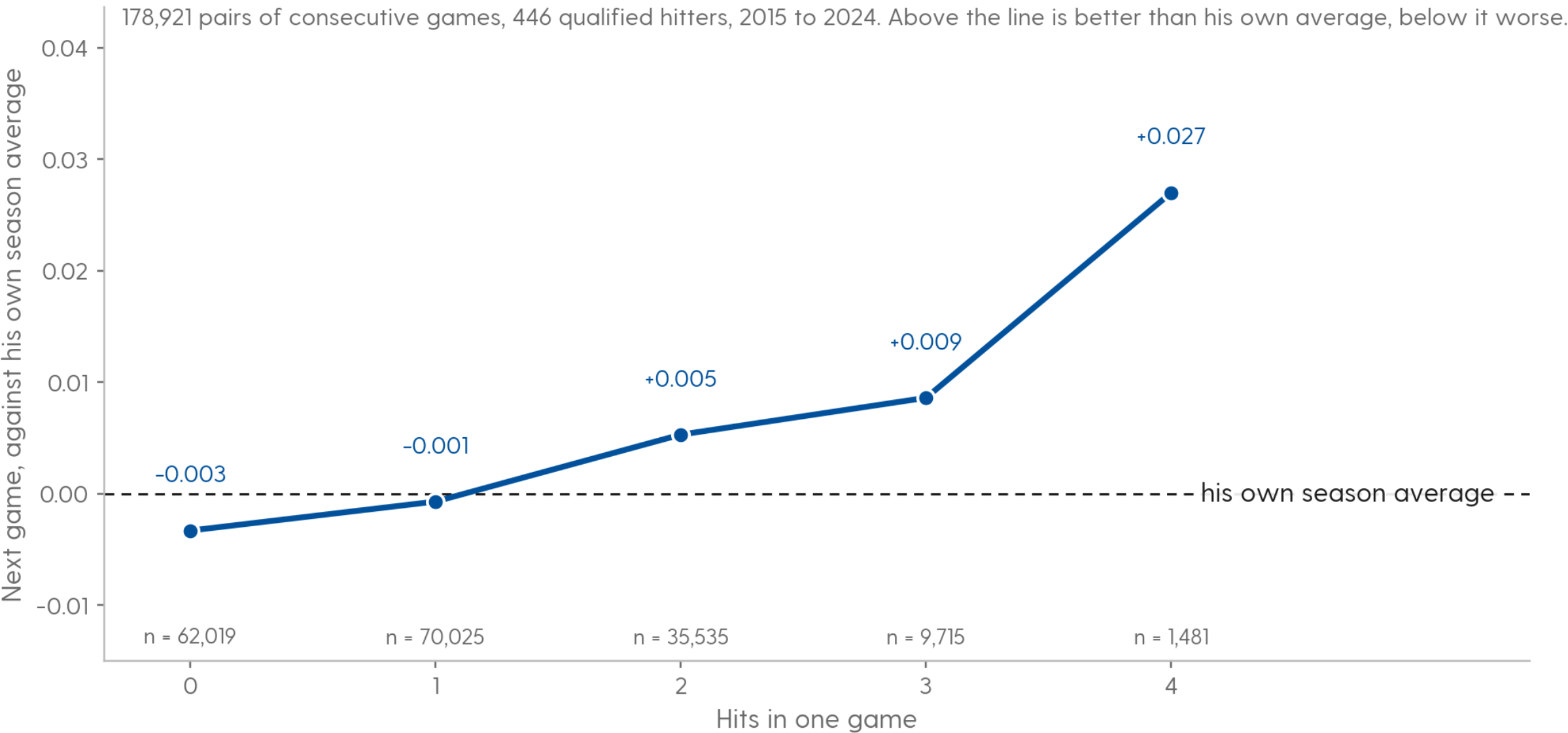
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POOLED ACROSS HITTERS



Pooled, a two-hit game is followed by more hits. Good hitters have more two-hit games and more hits the next day.

EACH HITTER AGAINST HIS OWN AVERAGE



C. Against his own average, a hitter moves by hundredths of a hit. Statistically significant, but not clinically significant: too small to act on.



Across 2024 pitchers, hits allowed and walks allowed correlate at 0.80. Expressed per nine innings, the same two things correlate at 0.03. What happened?

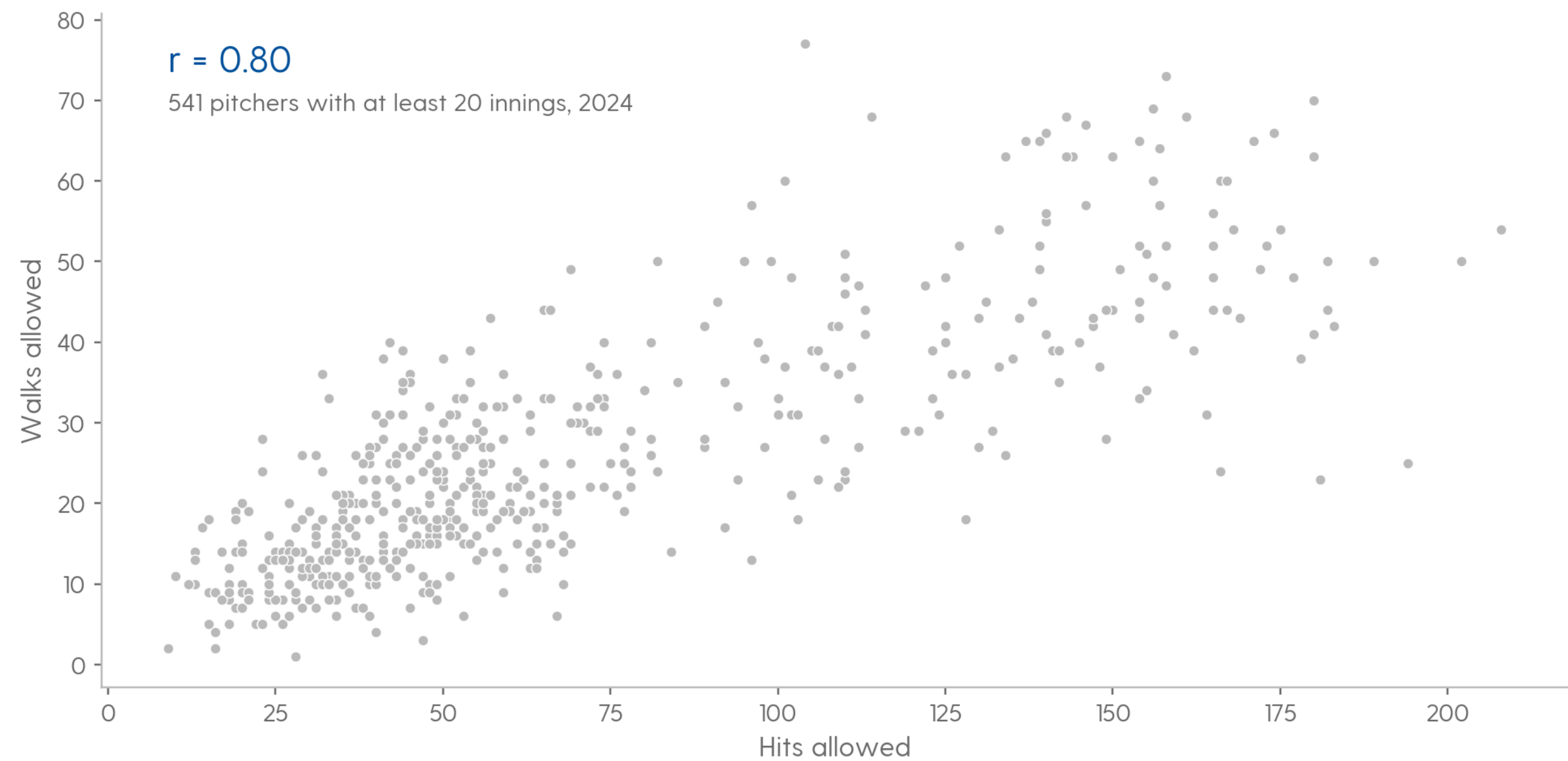
- A** Innings pitched moves both counts, and dividing by innings takes it out.
- B** Walks cause hits, but only for pitchers with many innings.
- C** The per-nine version has too few pitchers to show the relationship.
- D** Rounding to one decimal destroyed the correlation.



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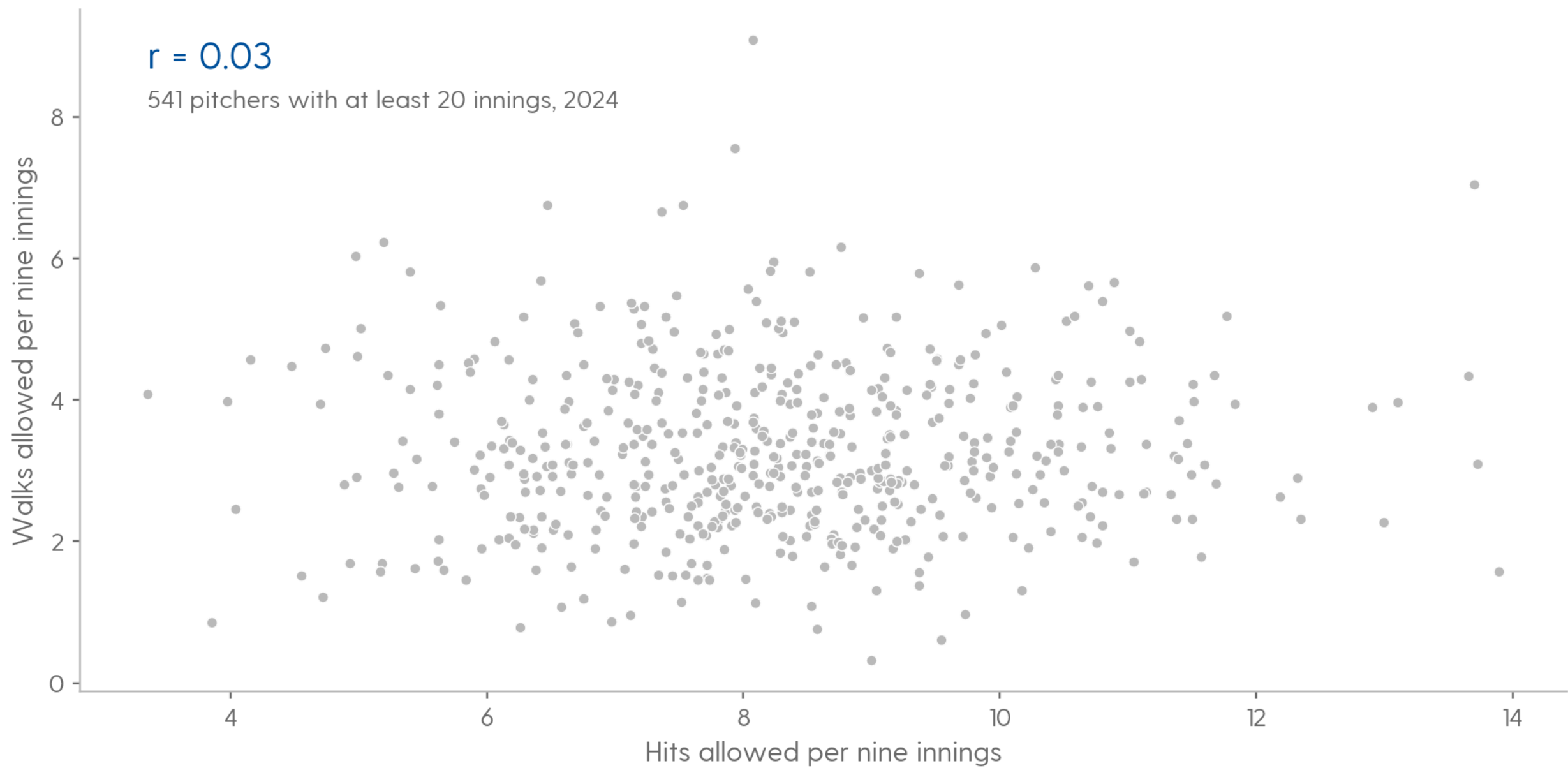
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HITS AND WALKS ALLOWED, 2024



Raw counts correlate at 0.80. Pitchers who pitch more innings allow more of both.

PER NINE INNINGS



A. Per nine innings, 0.03. Innings drove both counts.



Your agent checks a player's game log and reports meaningful autocorrelation: this game's value is correlated with the previous game's. Which is the standard next step in the model?

- A** Add the previous game's value as a predictor of this game's, a lagging variable.
- B** Average the season and model that instead.
- C** Report the result; autocorrelation is what streaky players look like.
- D** Drop every other game so consecutive rows are no longer adjacent.



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A. ADD LAST GAME AS A PREDICTOR.

When this game tracks the last one, put last game's value in the model as its own variable, a lagging variable. Then the other coefficients are read with the carryover already accounted for.

Positive autocorrelation: good games follow good games. Negative: a good game tends to follow a bad one.



On Thursday's ten games, the slope of margin on rush attempts came out at 0.90, $p = 0.076$, and a 95% interval from -0.12 to 1.92. The right reading is:

- A** The slope is a real-looking number, but ten games cannot distinguish it from zero, and the interval says the true slope could be anywhere from slightly negative to more than two points per attempt.
- B** The slope is significant because most of the interval is above zero.
- C** Each rush attempt is worth 0.90 points of margin; p is close enough to 0.05.
- D** Rushing has no effect on margin, because p is above 0.05.



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A. TEN GAMES CANNOT TELL 0.90 FROM ZERO.

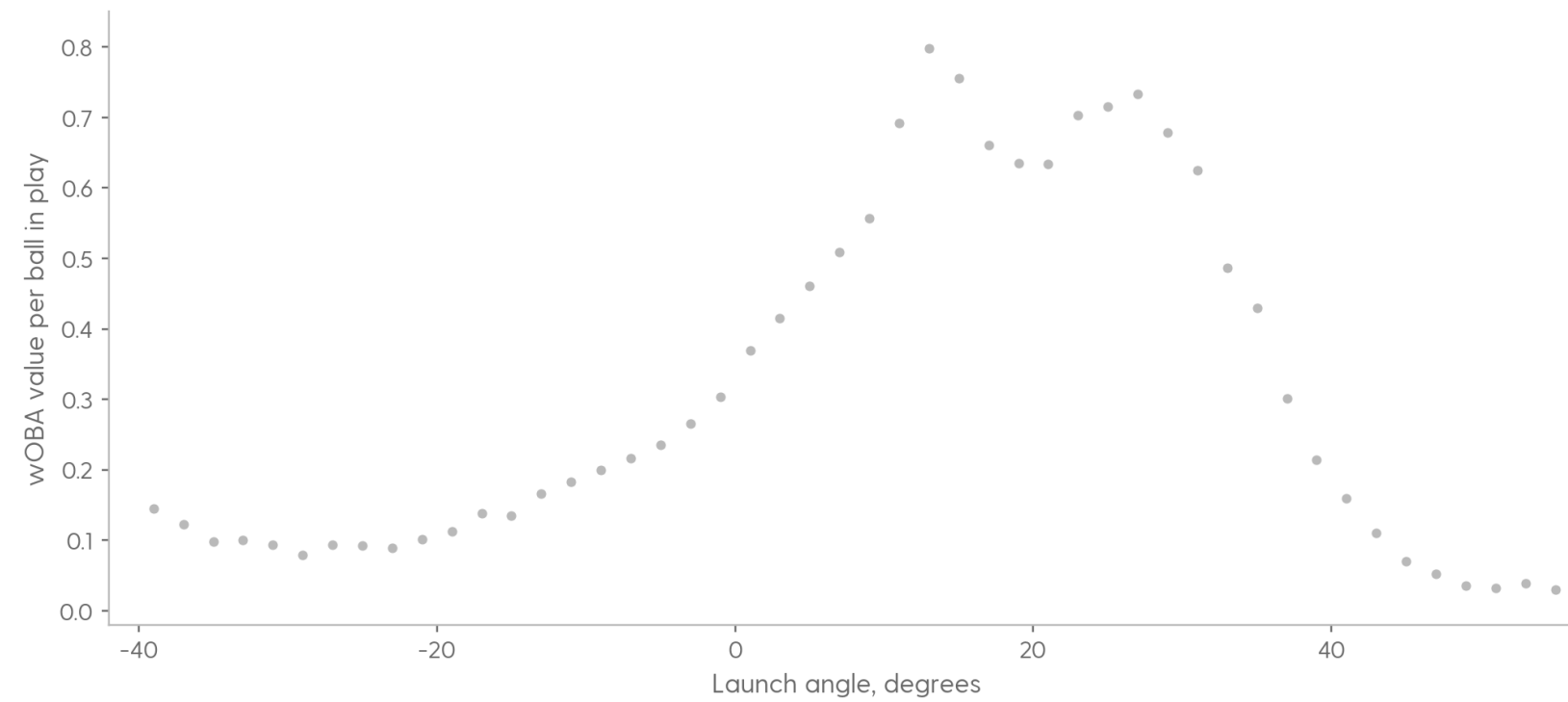
The 95% interval runs from -0.12 to 1.92. It includes zero, so the true slope could be nothing at all, or more than two points of margin per rush.

06

WEEK 6

Curves, interactions and dummies

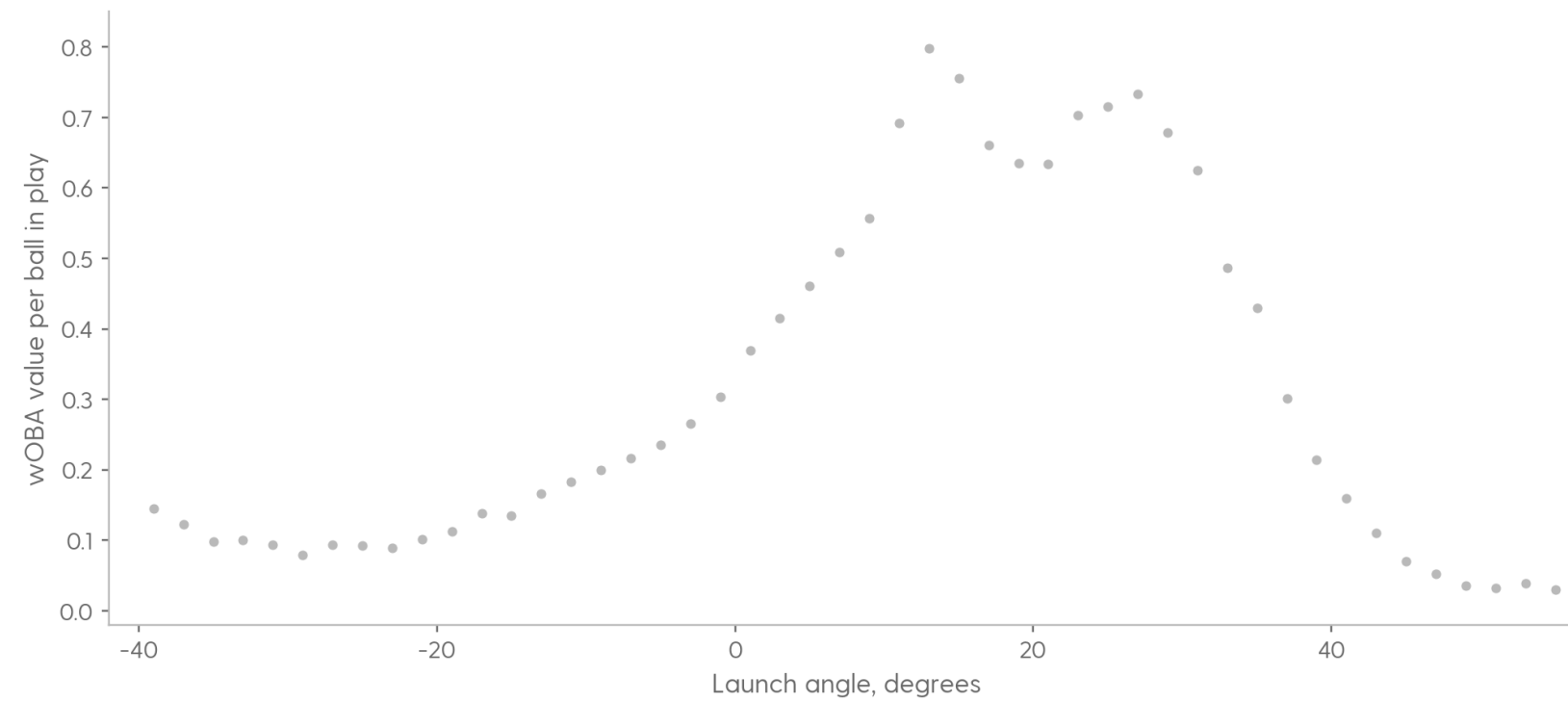
WEEK 6, QUESTION 1



When choosing the degree of a polynomial regression, you should consider:

- A** What maximizes the R^2 value.
- B** How many categories the variable has.
- C** The statistical significance of coefficients, scatterplot curvature, and logical shape.
- D** None of the above.

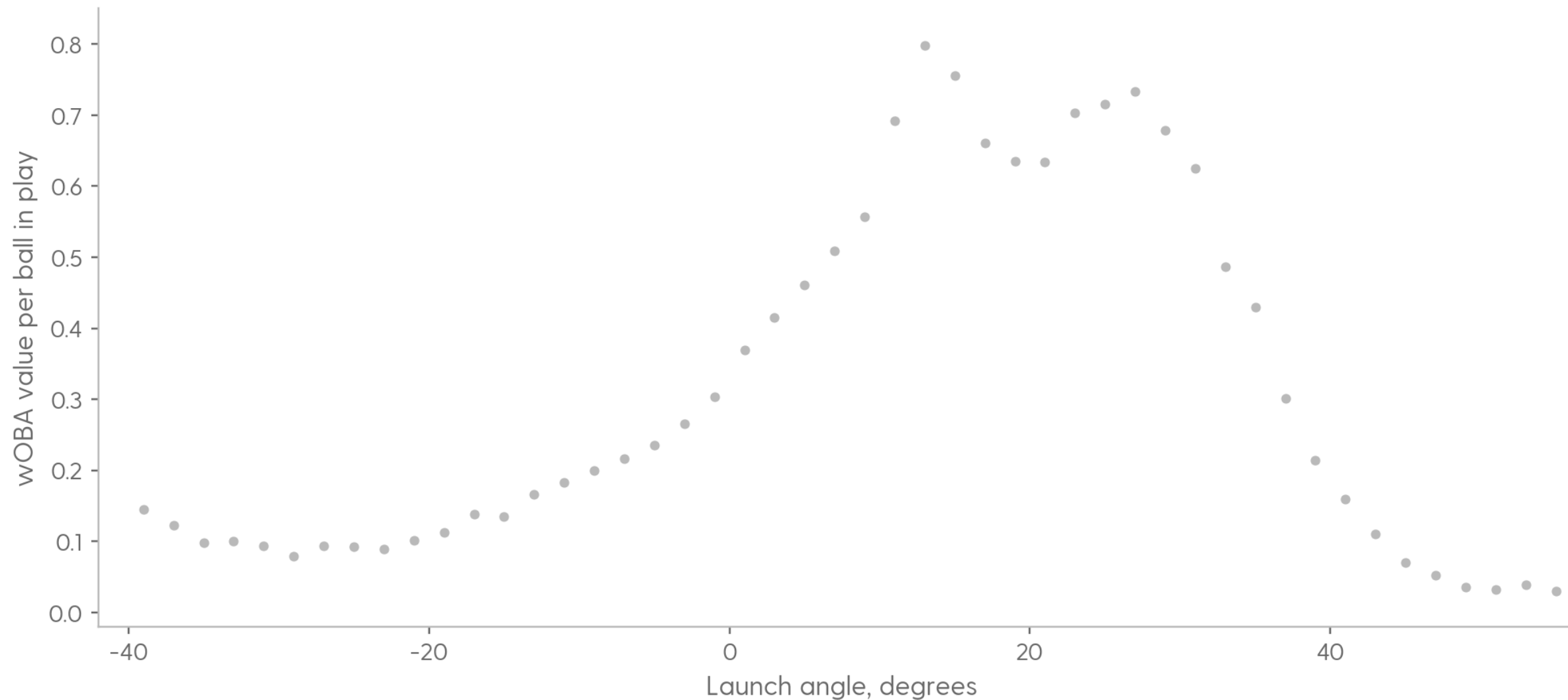
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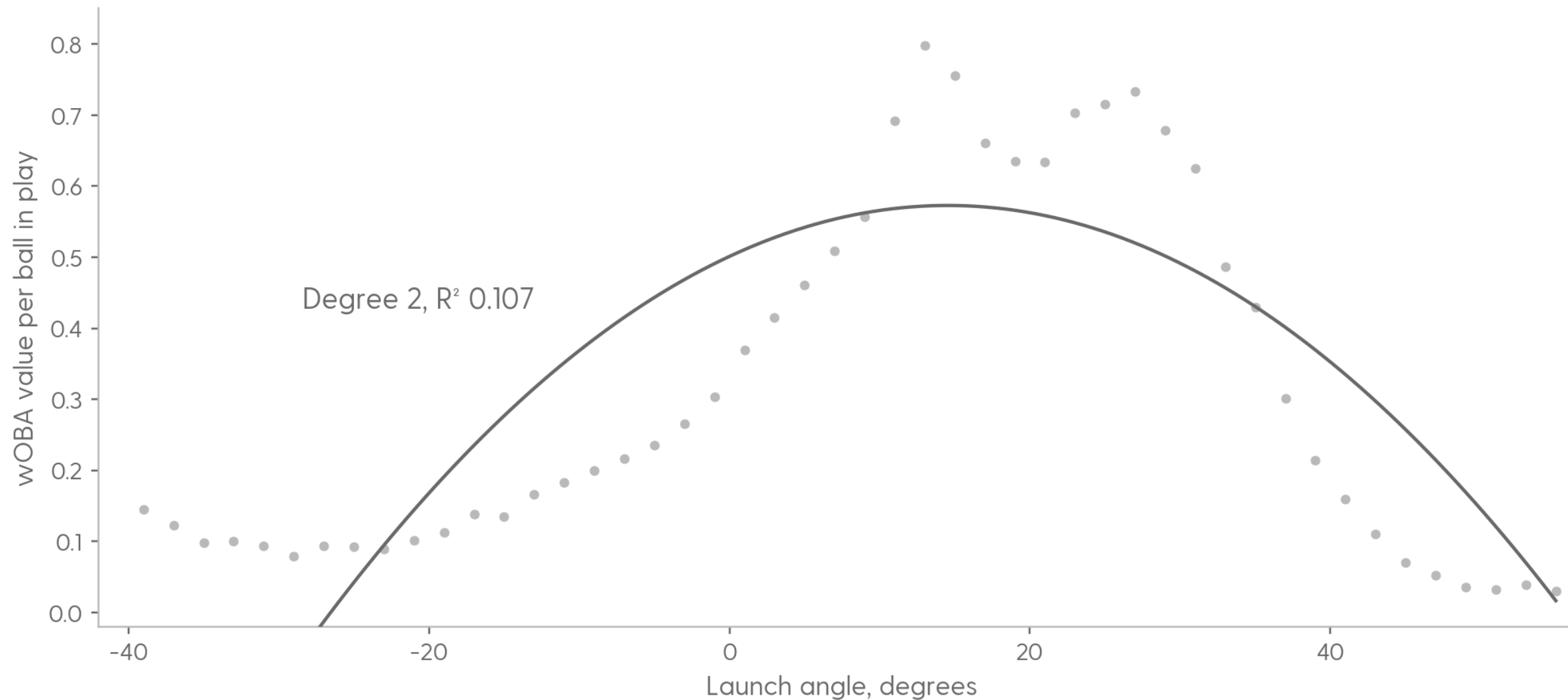
LAUNCH ANGLE AND WOBAs, 2025



Average wOBA value by launch angle. It falls into a valley near -30° , climbs to a peak near 22° , then drops off on pop-ups. That is two turns.

Statcast, 110,078 regular-season balls in play from -40° to 55° . Dots are 2-degree averages.

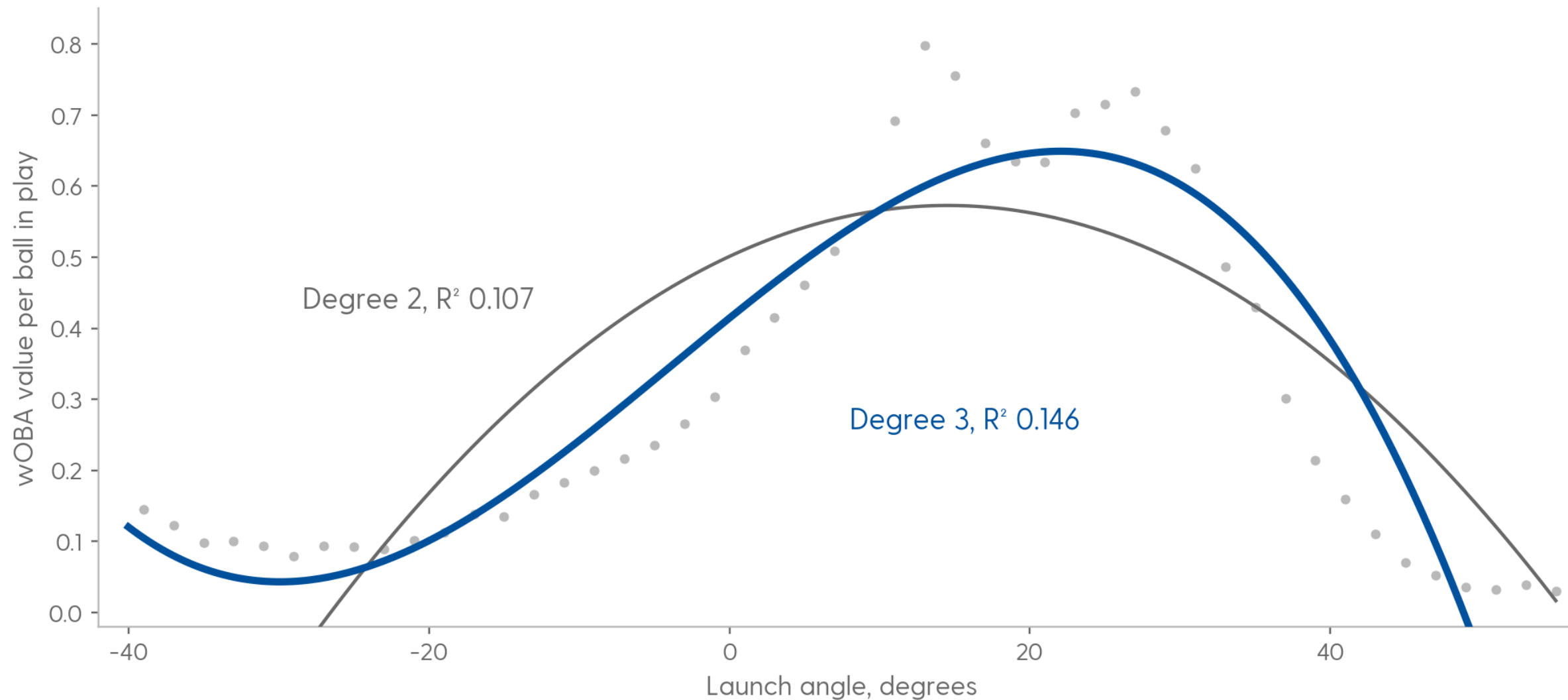
LAUNCH ANGLE AND WOBAS, 2025



Degree 2 can only turn once. It finds the peak but runs straight through the ground-ball valley. R^2 0.107.

Statcast, 110,078 regular-season balls in play from -40° to 55° . Dots are 2-degree averages.

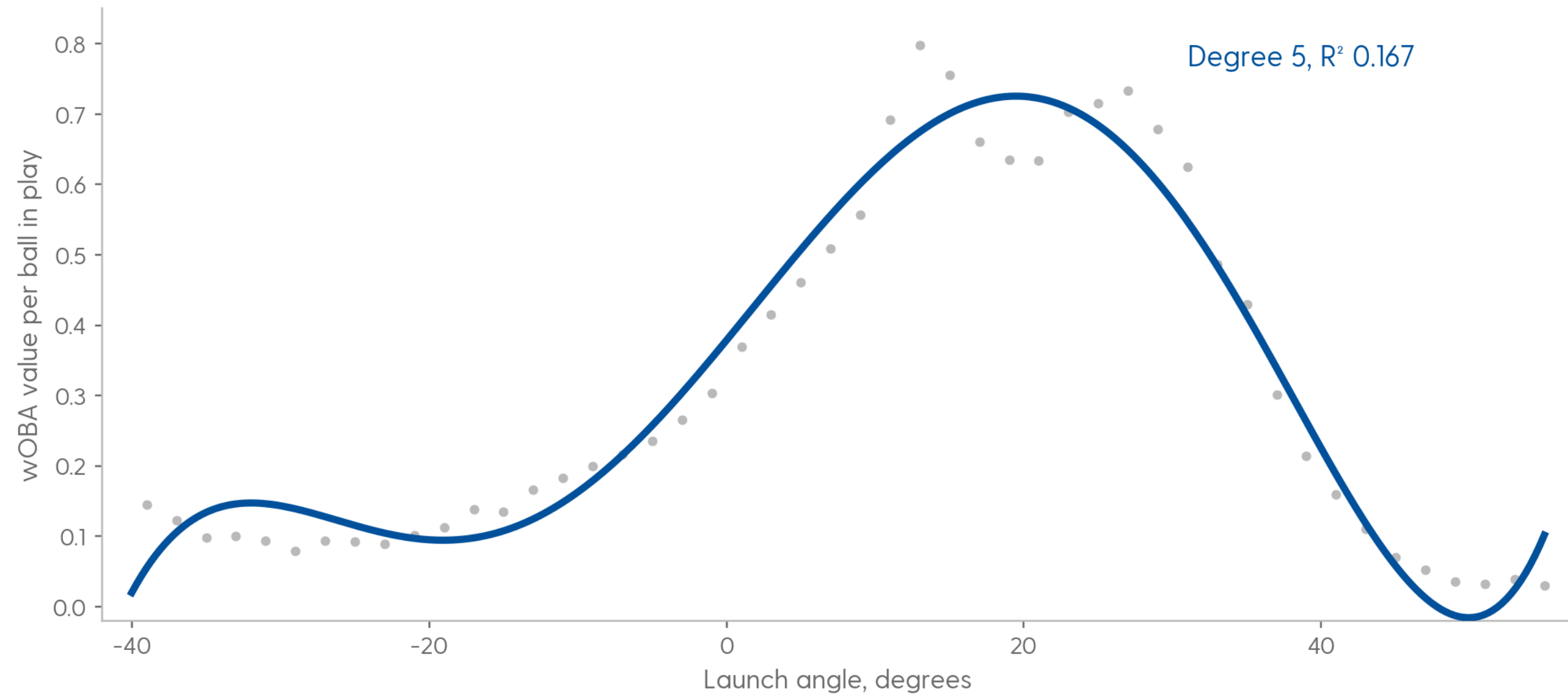
LAUNCH ANGLE AND WOBAs, 2025



Degree 3 can turn twice, and both turns have a baseball reason: worst near -30° (topped into the ground), best near 22° (line drives and fly balls). R^2 rises to 0.146.

Statcast, 110,078 regular-season balls in play from -40° to 55° . Dots are 2-degree averages.

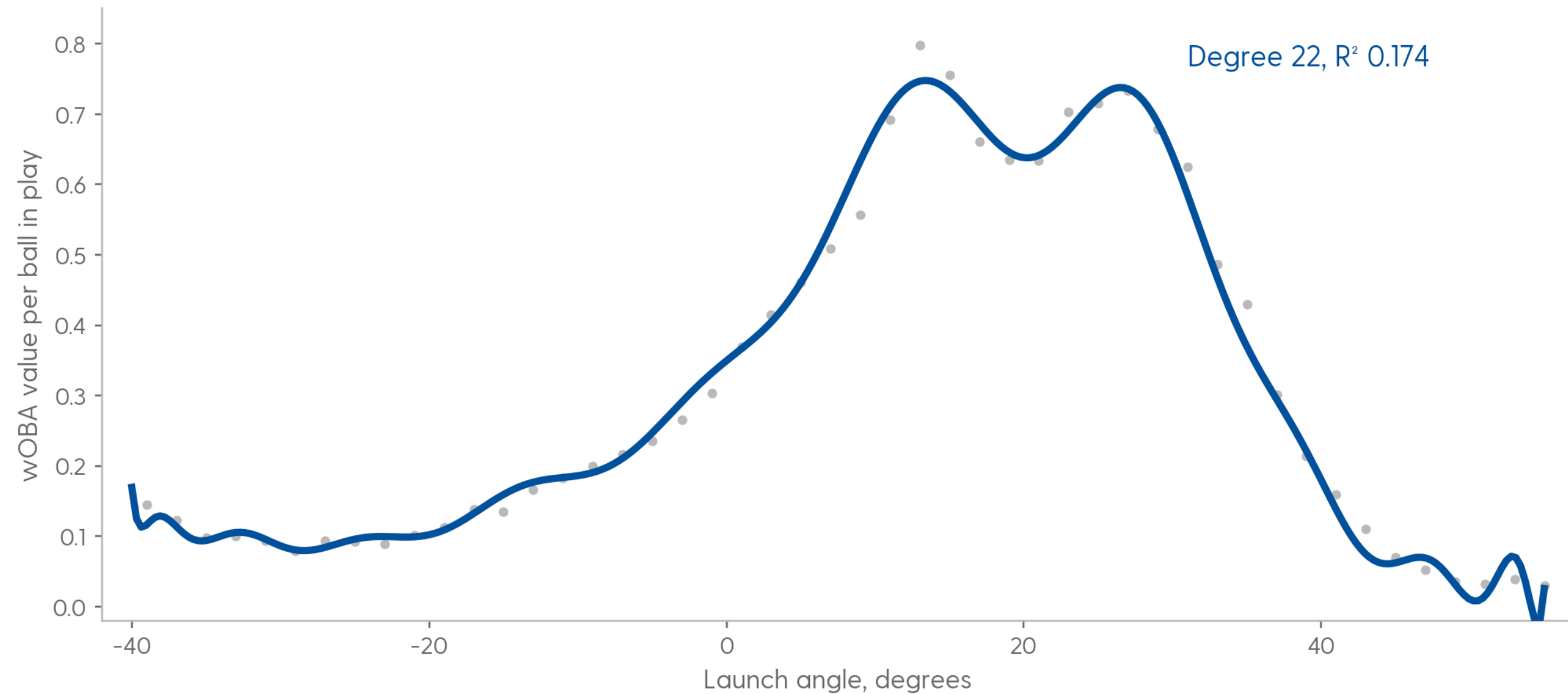
LAUNCH ANGLE AND WOBAs, 2025



C. Degree 3. Degree 5 adds only 0.022 and wiggles at the ends that no hitter would explain.

Statcast, 110,078 regular-season balls in play from -40° to 55°. Dots are 2-degree averages.

LAUNCH ANGLE AND WOB, 2025



The dip near 20° is real: about 1,800 balls in each degree, and the drop is ten times its standard error. Degree 22 is the first fit to catch it, and it holds up on new data (R^2 0.175 on a held-out half, vs 0.168 for degree 5). The price: 22 terms no one can read, and wiggles at the ends that dip below zero wOBA, which can't happen.

SPAX 402 · Week 7

Statcast, 110,078 regular-season balls in play from -40° to 55°. Dots are 2-degree averages.



In class, we predicted Points per Game as a function of average minutes on the ice, using the model $Y = 0.0905 - 0.0201T + 0.00280T^2$. What does the positive coefficient of 0.00280 next to T^2 tell us?

- A** The difference in points per game between players playing 10 & 11 minutes per game is predicted to be the same as players scoring 20 & 21 minutes per game.
- B** The difference in points per game between players playing 10 & 11 minutes per game is predicted to be larger than players scoring 20 & 21 minutes per game.
- C** The difference in points per game between players playing 10 & 11 minutes per game is predicted to be smaller than players scoring 20 & 21 minutes per game.



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C. THE 21ST MINUTE IS WORTH MORE THAN TWICE THE 11TH.

10 TO 11 MINUTES

3.1 points over an 80-game season.

20 TO 21 MINUTES

7.6 points over an 80-game season.



Among 2023-24 NHL forwards, those who averaged 18 or more minutes a game scored 2.8 points per 60 minutes of ice time. Those who averaged under 12 minutes scored 1.2 per 60 minutes. A coach wants to give his bench forwards more minutes, believing that the data shows that if he does so they'll "find their groove" and score at the higher rate. What is the main problem with that plan?

- A** The two groups have different numbers of players, so their rates can't be compared.
- B** Points per 60 minutes already adjusts for ice time, so the two groups are really scoring at the same rate.
- C** Coaches give the most minutes to their best scorers, so the gap mostly shows who earns minutes, not what minutes do.



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POINTS PER 60 MINUTES BY TIME ON ICE, 2023-24

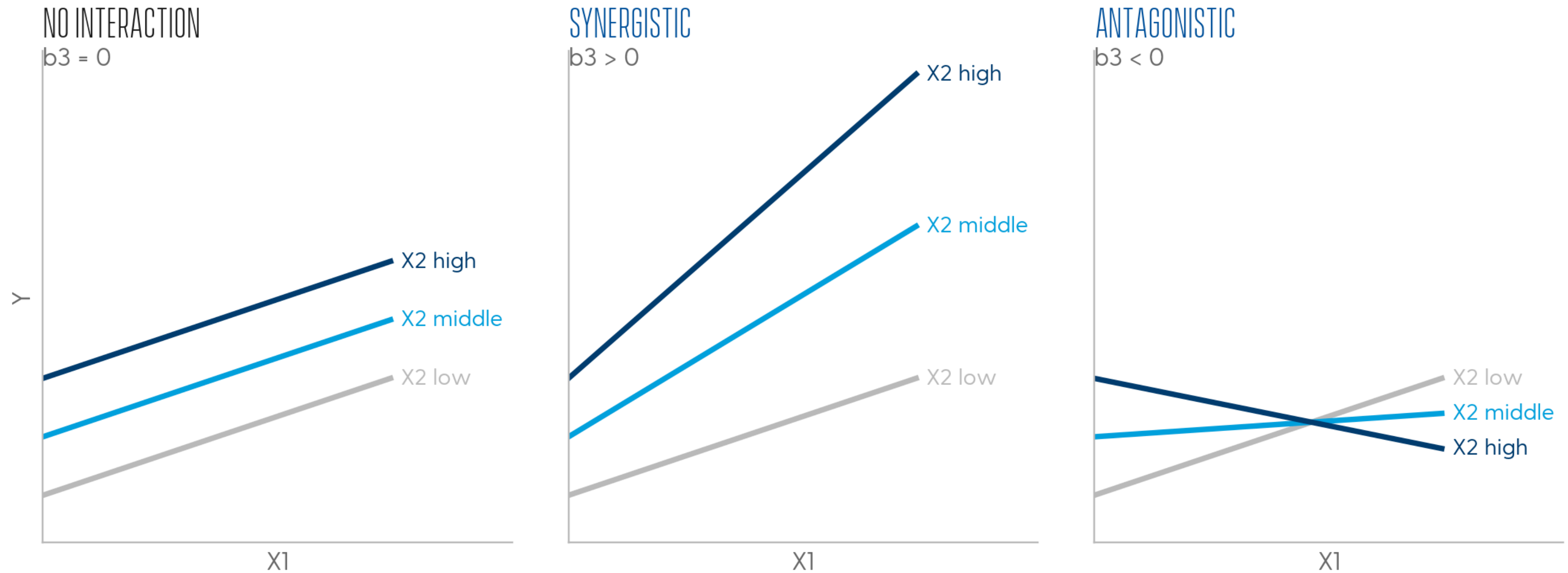
MINUTES PER GAME	FORWARDS	POINTS PER 60 MINUTES
Under 12	73	1.2
12 to 15	113	1.5
15 to 18	125	2.0
18 or more	95	2.8

C. Selection. Coaches hand out minutes on ability, so this table sorts players by how good they are. It cannot say what more minutes would do for a bench player.

? DOES A BIG ARM MAKE A FAST RECEIVER WORTH MORE, OR LESS?

INTERACTIONS

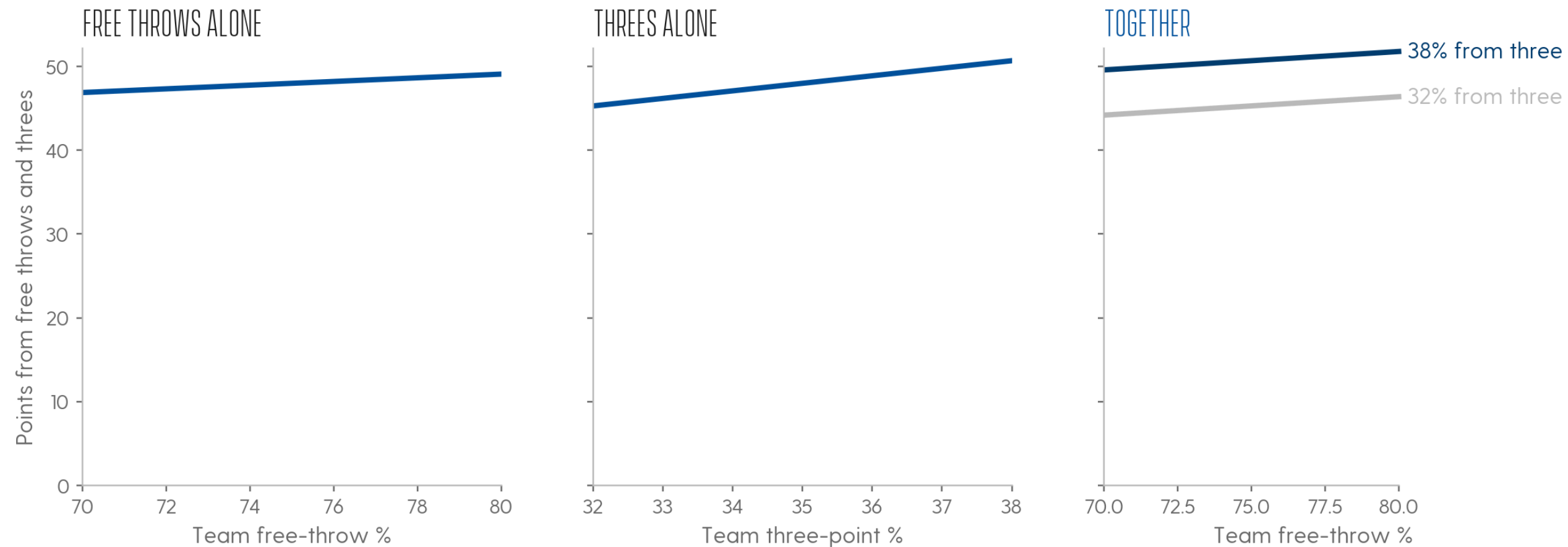
THREE SHAPES



Parallel lines: no interaction. Lines fan out: synergistic. Lines close in: antagonistic.

NO INTERACTION: FREE THROWS AND THREES

$$\text{Points} = a + b_1 \cdot \text{FT\%} + b_2 \cdot \text{3P\%} \quad (\text{no FT\%} \times \text{3P\% term})$$

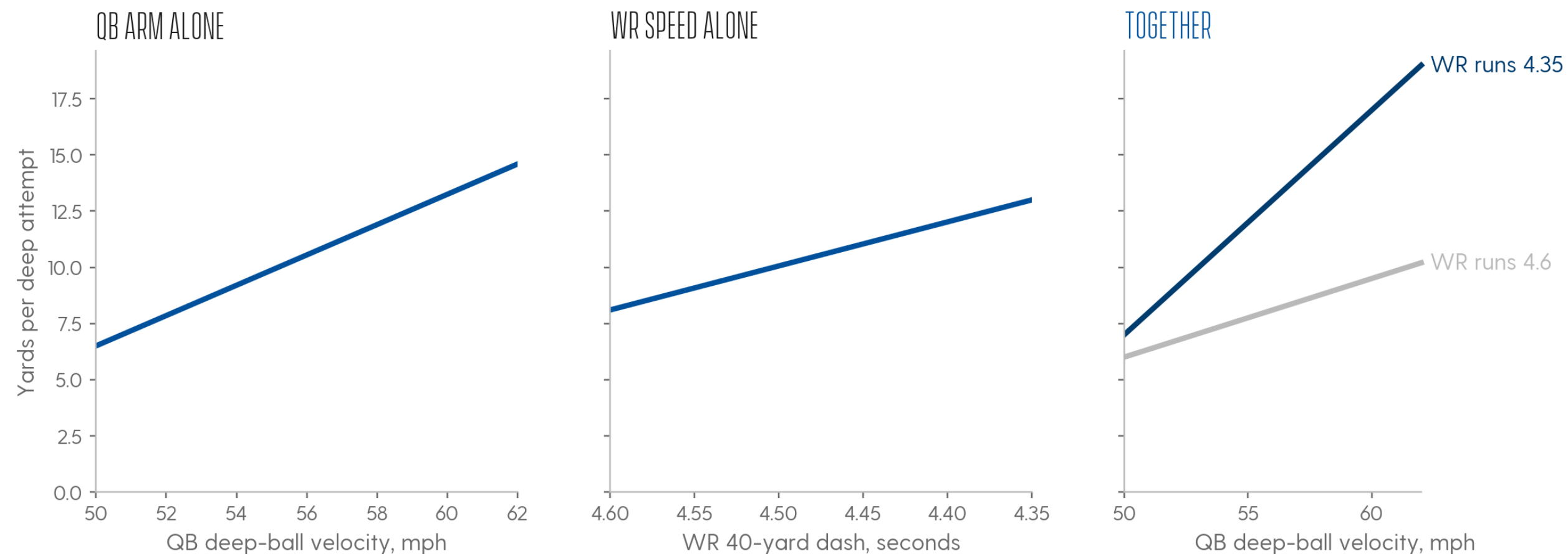


At 22 free throws and 30 threes a game, 10 more points of free-throw % adds about 2 points and 6 more points of three-point % adds about 5. Together the lines stay parallel: the free throws are worth the same 2 points however the threes are falling.

Illustrative numbers.

SYNERGISTIC: ARM AND SPEED

Yards = a + b1·Arm + b2·Speed + **b3·Arm × Speed**

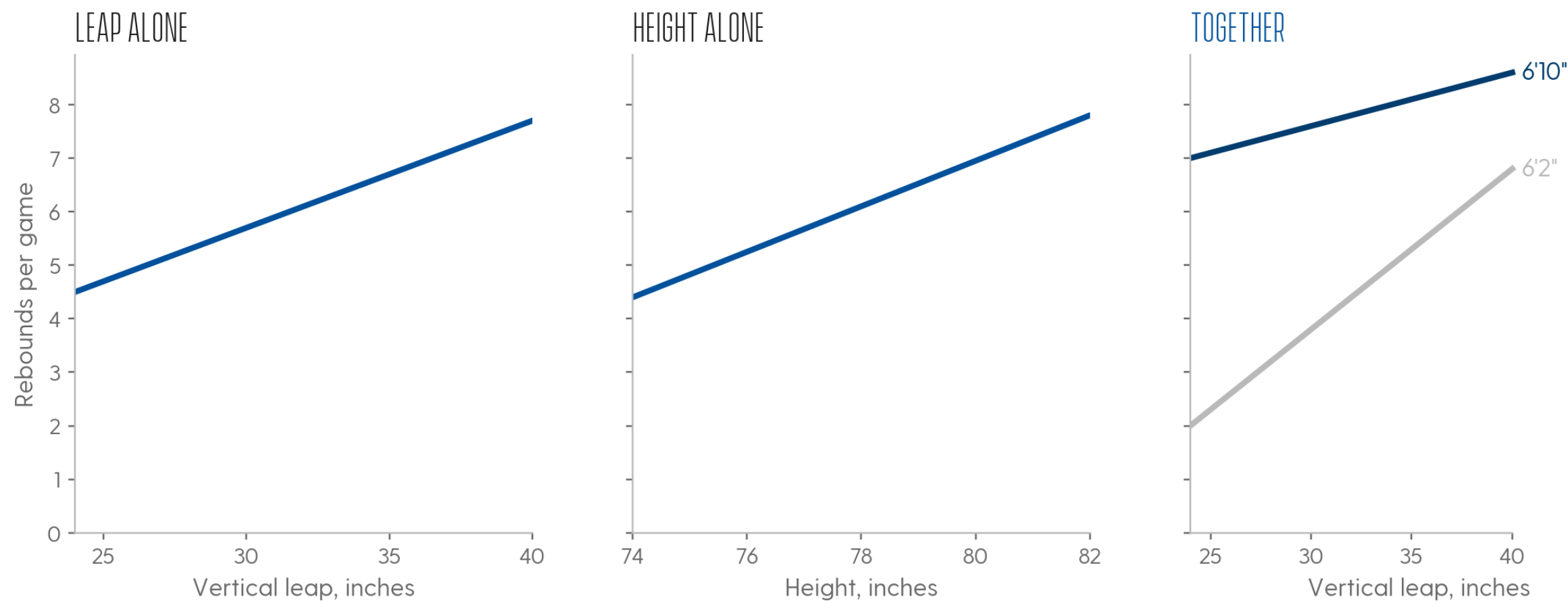


A bigger arm and a faster receiver each add yards on their own. Together the lines fan out: going from 50 to 62 mph is worth about 4 yards per deep attempt to a 4.6 receiver and 12 to a 4.35 receiver.

Illustrative numbers.

ANTAGONISTIC: LEAP AND HEIGHT

Rebounds = a + b1·Leap + b2·Height - b3·Leap × Height



Leap and height each add rebounds on their own. Together the lines close in: 16 more inches of leap adds about 5 rebounds at 6'2" and under 2 at 6'10". The slope gets shallower; it does not turn down.

Illustrative numbers.

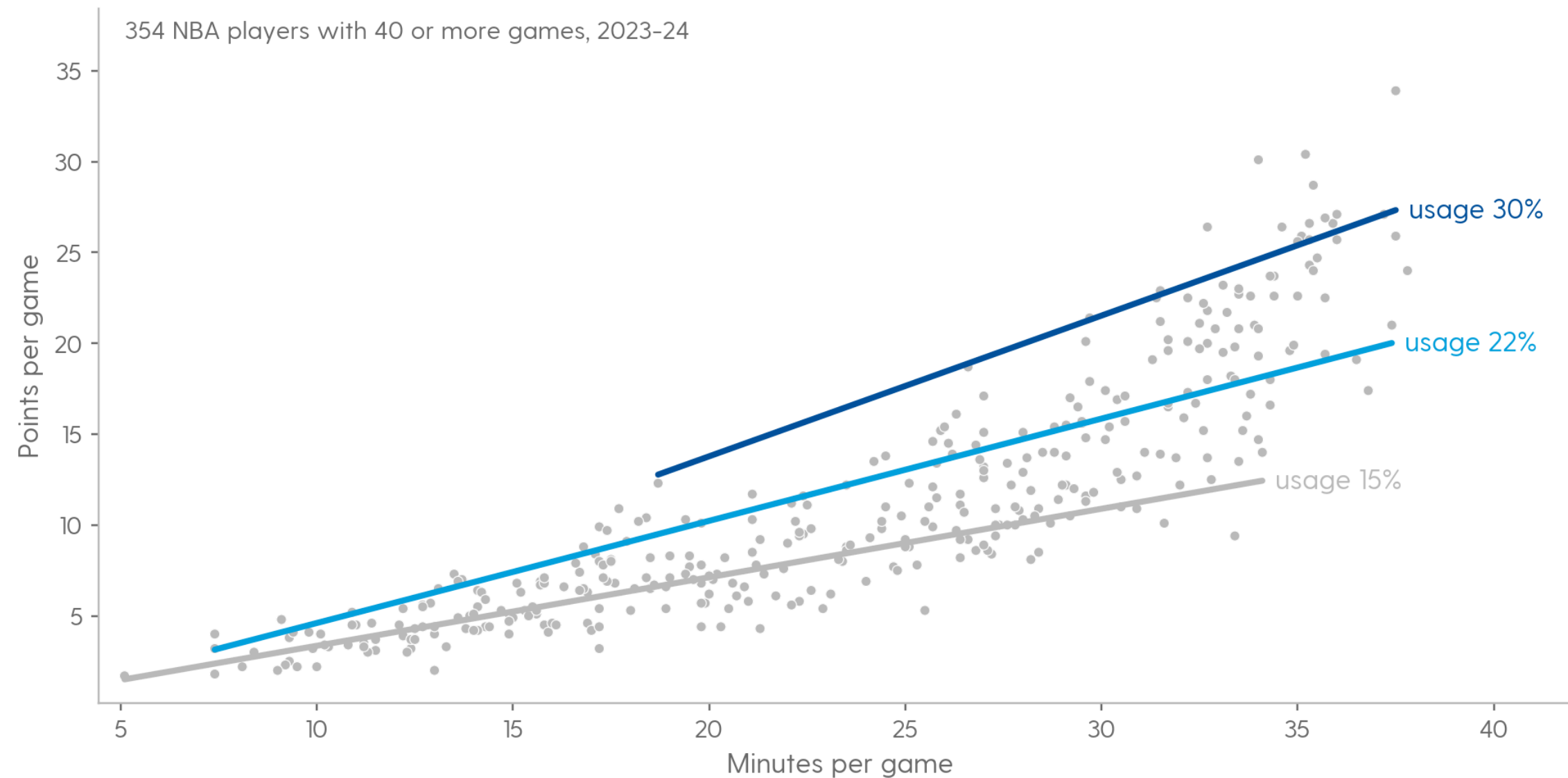
WHAT B3 MEASURES

$$Y = a + b_1X_1 + b_2X_2 + b_3X_1X_2$$

$$\text{Slope of } X_1 = b_1 + b_3 \times X_2$$

b_3 is how much each extra unit of X_2 changes what one unit of X_1 is worth.

MINUTES AND USAGE, 2023-24



Real data from Week 6. The lines fan out: a minute is worth more points to a high-usage player. Synergistic.

SYNERGISTIC PAIRS ARE TEAMMATES. ANTAGONISTIC PAIRS ARE DOING THE SAME JOB.

SYNERGISTIC, $B3 > 0$

Together they add more than the sum of each alone.

ANTAGONISTIC, $B3 < 0$

Together they add less than the sum of each alone.

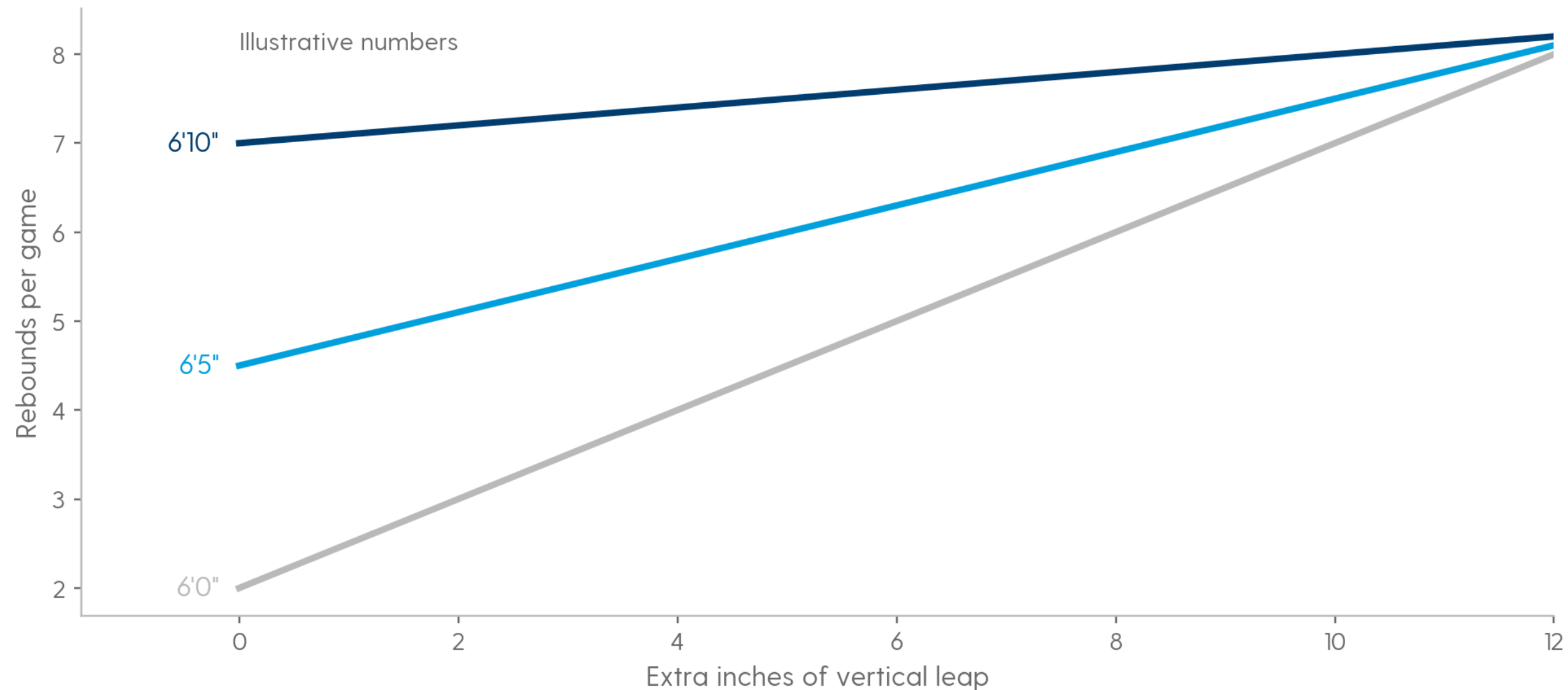
**? A RUNNING BACK'S SPEED AND HIS LINE'S RUN
BLOCKING, FOR RUSHING YARDS. SYNERGISTIC OR
ANTAGONISTIC?**

**SYNERGISTIC. BETTER BLOCKING OPENS
HOLES, AND SPEED IS WORTH MORE THROUGH
A HOLE.**

**? TWO SHOT BLOCKERS ON THE FLOOR AT ONCE, FOR
BLOCKED SHOTS. SYNERGISTIC OR ANTAGONISTIC?**

**ANTAGONISTIC. THEY DO THE SAME JOB, SO
THE SECOND ONE ADDS LESS WITH THE FIRST
ALREADY THERE.**

ANTAGONISTIC DOES NOT MEAN BAD



b3 is negative, and the tallest line is still on top. Having both adds less than the two effects added up. It says nothing about whether either one helps on its own.



We want to create dummy variables for each division in the NFL. In the NFL, there are the following divisions: NFC East, NFC West, NFC North, NFC South, AFC East, AFC West, AFC North, and AFC South. How many dummy variables should we create?

- A 1
- B 7
- C 2
- D 8



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- D 8

NFL DIVISIONS

DIVISION	AFC N	AFC S	AFC W	NFC E	NFC N	NFC S	NFC W
AFC East	0	0	0	0	0	0	0
AFC North	1	0	0	0	0	0	0
AFC South	0	1	0	0	0	0	0
AFC West	0	0	1	0	0	0	0
NFC East	0	0	0	1	0	0	0
NFC North	0	0	0	0	1	0	0
NFC South	0	0	0	0	0	1	0
NFC West	0	0	0	0	0	0	1

B. 7. Each one is a gap from AFC East, the reference.